

CBSE CLASS X
Mathematics Standard (041)

QUESTION PAPER

From previous CBSE Board Exam questions

Code: CYDFDR**Questions: 69****Maximum Marks: 116****Generated: 2026-06-15 13:05****SELECTIONS USED**

Source	Previous-year board
Subject	Mathematics
Lessons	Probability
Year	2022-2026
Questions selected	69

Q1. § 13.4 Median of Grouped Data**[4]**

BINGO is game of chance. The host has 75 balls numbered 1 through 75. Each player has a BINGO card with some numbers written on it. The participant cancels the number on the card when called out a number written on the ball selected at random. Whosoever cancels all the numbers on his/her card, says BINGO and wins the game. The table given below, shows the data of one such game where 48 balls were used before Tara said 'BINGO'.

Based on the above information, answer the following :

- (i) Write the median class. [1]
- (ii) When first ball was picked up, what was the probability of calling out an even number ? [1]
- (iii) Find median of the given data. [2]

Previously asked in: 2024 30/1/1 Q37

◆ **Statistics**

13.4 Median of Grouped Data

14.1 Probability — A Theoretical Approach

Q2. § Chapter Opening and Epigraph**[2]**

In a pack of 52 playing cards one card is lost. From the remaining cards, a card is drawn at random. Find the probability that the drawn card is queen of heart, if the lost card is a black card.

Previously asked in: 2024 30/1/1 Q22

◆ **Probability**

14.1 Probability — A Theoretical Approach

Q3. § Chapter Opening and Epigraph § 14.1 Probability — A Theoretical Approach § 14.2 Summary**[1]**

Two dice are rolled together. The probability of getting sum of numbers on the two dice as 2, 3 or 5, is :

- (a) $\frac{7}{36}$
- (b) $\frac{11}{36}$
- (c) $\frac{5}{36}$
- (d) $\frac{4}{9}$

Previously asked in: 2024 30/1/1 Q16

◆ **Probability**

14.1 Probability — A Theoretical Approach

Q4. § Chapter Opening and Epigraph § 14.1 Probability — A Theoretical Approach § 14.2 Summary**[1]**

From the data 1, 4, 7, 9, 16, 21, 25, if all the even numbers are removed, then the probability of getting at random a prime number from the remaining is :

- (a) $\frac{2}{5}$
- (b) $\frac{1}{5}$
- (c) $\frac{1}{7}$
- (d) $\frac{2}{7}$

Previously asked in: 2024 30/1/1 Q8

◆ **Probability** 14.1 Probability — A Theoretical Approach

Q5. § Chapter Opening and Epigraph**[1]**

If the probability of a player winning a game is 0.79, then the probability of his losing the same game is :

- (a) 1.79
- (b) 0.31
- (c) 0.21%
- (d) 0.21

Previously asked in: 2024 30/1/1 Q2

◆ **Probability** 14.1 Probability — A Theoretical Approach

Q6. § 14.2 Summary**[2]**

A bag contains 25 balls. Some of them are yellow and others are green. One ball is drawn at random. If probability of getting a green ball is $\frac{3}{5}$, then find the number of yellow balls.

Previously asked in: 2026 30/5/1 Q25

◆ **Probability** 14.1 Probability — A Theoretical Approach

Q7. § Chapter Opening and Epigraph § 14.2 Summary**[1]**

Meena calculates that the probability of her winning the first prize in a lottery is 0.08. If total 800 tickets were sold, the number of tickets bought by her, is

- (A) 64
- (B) 640
- (C) 100
- (D) 10

Previously asked in: 2026 30/5/1 Q16

◆ **Probability** 14.1 Probability — A Theoretical Approach

Q8. § Chapter Opening and Epigraph § 14.1 Probability — A Theoretical Approach**[1]**

A card is drawn at random from a well shuffled deck of 52 playing cards. The probability that it is either a ten or a king is

- (A) $\frac{1}{26}$
- (B) $\frac{2}{13}$
- (C) $\frac{1}{13}$
- (D) $\frac{8}{26}$

Previously asked in: 2026 30/5/1 Q11

◆ **Probability** 14.1 Probability — A Theoretical Approach

Q9. § 14.2 Summary**[1]**

Which of the following can not be the probability of an event?

- (A) $\frac{39}{100}$
 (B) $\frac{0.001}{20}$
 (C) $\frac{10}{0.2}$
 (D) 10%

Previously asked in: 2026 30/5/1 Q10

◆ Probability

14.1 Probability — A Theoretical Approach

Q10. § Chapter Opening and Epigraph § 14.1 Probability — A Theoretical Approach**[4]**

"Eight Ball" is a game played on a pool table with 15 balls numbered 1 to 15 and a "cue ball" that is solid and white. Of the 15 numbered balls, eight are solid (non-white) coloured and numbered 1 to 8 and seven are striped balls numbered 9 to 15.

The 15 numbered pool balls (no cue ball) are placed in a large bowl and mixed, then one ball is drawn out at random.

Based on the above information, answer the following questions:

- (i) What is the probability that the drawn ball bears number 8? [1]
 (ii) What is the probability that the drawn ball bears an even number? [2]
 (iii) What is the probability that the drawn ball is a solid coloured and bears an even number? [1]

Previously asked in: 2023 30/6/1 Q38

◆ Probability

14.1 Probability — A Theoretical Approach

Q11. § Chapter Opening and Epigraph § 14.1 Probability — A Theoretical Approach**[1]**

A card is drawn at random from a well shuffled deck of 52 playing cards. The probability of getting a face card is

- A $\frac{1}{2}$
 B $\frac{3}{13}$
 C $\frac{4}{13}$
 D $\frac{1}{13}$

Previously asked in: 2023 30/6/1 Q15

◆ Probability

14.1 Probability — A Theoretical Approach

Q12. § Chapter Opening and Epigraph**[1]**

A bag contains 5 pink, 8 blue and 7 yellow balls. One ball is drawn at random from the bag. What is the probability of getting neither a blue nor a pink ball?

- A $\frac{1}{4}$
 B $\frac{2}{5}$
 C $\frac{7}{20}$
 D $\frac{13}{20}$

Previously asked in: 2023 30/6/1 Q11

◆ Probability

14.1 Probability — A Theoretical Approach

Q13. § Chapter Opening and Epigraph**[2]**

A bag contains 4 red, 3 blue and 2 yellow balls. One ball is drawn at random from the bag. Find the probability that drawn ball is (i) red (ii) yellow.

Previously asked in: 2023 30/1/1 Q25

◆ Probability

14.1 Probability — A Theoretical Approach

Q14. § Chapter Opening and Epigraph § 14.1 Probability — A Theoretical Approach § 14.2 Summary**[1]**

In a group of 20 people, 5 can't swim. If one person is selected at random, then the probability that he/she can swim, is

- A $\frac{3}{4}$
 B $\frac{1}{3}$
 C 1
 D $\frac{1}{4}$

Previously asked in: 2023 30/1/1 Q18

◆ Probability

14.1 Probability — A Theoretical Approach

Q15. § Chapter Opening and Epigraph § 14.2 Summary**[1]**

A girl calculates that the probability of her winning the first prize in a lottery is 0.08. If 6000 tickets are sold, how many tickets has she bought ?

- A 40
 B 240
 C 480
 D 750

Previously asked in: 2023 30/1/1 Q17

◆ Probability

14.1 Probability — A Theoretical Approach

Q16. § Chapter Opening and Epigraph § 14.2 Summary**[1]**

Probability of happening of an event is denoted by p and probability of non-happening of the event is denoted by q. Relation between p and q is

- A $p + q = 1$
 B $p = 1, q = 1$
 C $p = q - 1$
 D $p + q + 1 = 0$

Previously asked in: 2023 30/1/1 Q16

◆ Probability

14.1 Probability — A Theoretical Approach

Q17. § Chapter Opening and Epigraph**[2]**

The number of red balls in a bag is three more than the number of black balls. If the probability of drawing a red ball at random from the given bag is $\frac{12}{23}$, find the total number of balls in the given bag.

Previously asked in: 2025 30/6/1 Q25

◆ Probability

14.1 Probability — A Theoretical Approach

Q18. § 14.2 Summary**[1]**

Directions: A statement of Assertion (A) is followed by a statement of Reason (R). Choose the correct option.

In an experiment of throwing a die,

Assertion (A): Event E_1 : getting a number less than 3 and Event E_2 : getting a number greater than 3 are complementary events.

Reason (R): If two events E and F are complementary events, then $P(E) + P(F) = 1$.

- A Both Assertion (A) and Reason (R) are true and Reason (R) is correct explanation of Assertion (A).
 B Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of Assertion (A).
 C Assertion (A) is true, but Reason (R) is false.
 D Assertion (A) is false, but Reason (R) is true.

Previously asked in: 2025 30/6/1 Q20

◆ Probability

14.1 Probability — A Theoretical Approach

Q19. § 14.1 Probability — A Theoretical Approach**[1]**

Two coins are tossed simultaneously. The probability of getting atleast one head is

- A $\frac{1}{4}$
- B $\frac{1}{2}$
- C $\frac{3}{4}$
- D 1

Previously asked in: 2025 30/6/1 Q8

♦ Probability

14.1 Probability — A Theoretical Approach

Q20. § Chapter Opening and Epigraph § 14.1 Probability — A Theoretical Approach**[2]**

While shuffling a pack of 52 cards, one card was accidentally dropped. Find the probability that the dropped card

- (i) is not a face card.
- (ii) is a black king.

Previously asked in: 2025 30/5/1 Q25

♦ Probability

14.1 Probability — A Theoretical Approach

Q21. § 14.1 Probability — A Theoretical Approach § 14.2 Summary**[1]**

The probability of getting a composite number greater than 3 on throwing a die is

- A $\frac{1}{6}$
- B $\frac{1}{3}$
- C $\frac{1}{2}$
- D $\frac{2}{3}$

Previously asked in: 2025 30/5/1 Q18

♦ Probability

14.1 Probability — A Theoretical Approach

Q22. § Chapter Opening and Epigraph**[1]**

A bag contains red balls and black balls in the ratio 3 : 7. A ball is drawn at random. The probability that ball so drawn is black in colour, is

- A $\frac{3}{7}$
- B 0.3
- C 0.7
- D $\frac{1}{7}$

Previously asked in: 2025 30/5/1 Q7

♦ Probability

14.1 Probability — A Theoretical Approach

Q23.**[2]**

Two friends Anil and Ashraf were born in the December month in the year 2010. Find the probability that :

- (i) they share same date of birth. [1]
- (ii) they have different dates of birth. [1]

Previously asked in: 2025 30/4/1 Q25

♦ Probability

14.1 Probability — A Theoretical Approach

Q24. § 14.1 Probability — A Theoretical Approach

[1]

A pair of dice is thrown. The probability that sum of numbers appearing on top faces is at most 10 is :

- (a) $\frac{1}{11}$
- (b) $\frac{10}{11}$
- (c) $\frac{5}{6}$
- (d) $\frac{11}{12}$

Previously asked in: 2025 30/4/1 Q18

◆ Probability

14.1 Probability — A Theoretical Approach

Q25. § Chapter Opening and Epigraph § 14.2 Summary

[1]

The number of red balls in a bag is 10 more than the number of black balls. If the probability of drawing a red ball at random from this bag is $\frac{3}{5}$, then the total number of balls in the bag is :

- (a) 50
- (b) 60
- (c) 80
- (d) 40

Previously asked in: 2025 30/4/1 Q5

◆ Probability

14.1 Probability — A Theoretical Approach

Q26. § Chapter Opening and Epigraph § 14.1 Probability — A Theoretical Approach § 14.2 Summary

[4]

A group of friends wanted to play cards with two identical packs together. While shuffling the cards, three cards are dropped. Rest of the cards are shuffled and one card is drawn at random. Assuming that the dropped cards were a queen of hearts, a ten of spades and an ace of clubs, answer the following questions:

Assuming that the dropped cards were a queen of hearts, a ten of spades and an ace of clubs, answer the following questions:

- (i) Find the probability that the drawn card is a face card. [1]
- (ii) Find the probability that the drawn card is either a king or a queen. [1]
- (iii) Do you think that the probability of getting a queen was higher if none of the cards were dropped? Justify your answer. [2]

Previously asked in: 2026 30/3/1 Q37

◆ Probability

14.1 Probability — A Theoretical Approach

Probability with combined/modified decks of cards

Q27. § 14.1 Probability — A Theoretical Approach

[1]

Two dice are rolled together. The probability that the sum of the numbers obtained is divisible by 6, is :

- A $\frac{1}{6}$
- B $\frac{11}{36}$
- C $\frac{1}{12}$
- D $\frac{1}{4}$

Previously asked in: 2026 30/3/1 Q5

◆ Probability

14.1 Probability — A Theoretical Approach

Q28. § 14.1 Probability — A Theoretical Approach § 14.2 Summary

[3]

Two dice are thrown at the same time. Determine the probability that the (i) sum of the numbers on the two dice is 5, and (ii) difference of the numbers on the two dice is 3.

Previously asked in: 2026 30/2/1 Q31

◆ Probability

14.1 Probability — A Theoretical Approach

Q29. § Chapter Opening and Epigraph § 14.2 Summary**[1]**

The probability for a randomly selected number out of 1, 2, 3, 4, ..., 25 to be a composite number is :

- A $\frac{15}{25}$
- B $\frac{10}{25}$
- C $\frac{11}{25}$
- D $\frac{9}{25}$

Previously asked in: 2026 30/2/1 Q18

◆ **Probability**

14.1 Probability — A Theoretical Approach

Q30. § 14.1 Probability — A Theoretical Approach**[3]**

Two dice of different colours are thrown at the same time. Write down all the possible outcomes. What is the probability that :

- (i) same number appears on both the dice ?
- (ii) different number appears on both the dice ?

Previously asked in: 2026 30/1/1 Q31

◆ **Probability**

14.1 Probability — A Theoretical Approach

Q31. § 14.2 Summary**[1]**

Directions: Two statements are given, one labelled as Assertion (A) and the other labelled as Reason (R). Select the correct answer from the codes (a), (b), (c) and (d).

Assertion (A) : The probability that a leap year has 53 Mondays is $\frac{2}{7}$.

Reason (R) : The probability that a non-leap year has 53 Mondays is $\frac{5}{7}$.

- (a) Both, Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).
- (b) Both, Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).
- (c) Assertion (A) is true, but Reason (R) is false.
- (d) Assertion (A) is false, but Reason (R) is true.

Previously asked in: 2026 30/1/1 Q19

◆ **Probability**

14.1 Probability — A Theoretical Approach

Q32. § 14.1 Probability — A Theoretical Approach § 14.2 Summary**[1]**

A die is thrown once. Probability of getting a number other than 3 is :

- (a) $\frac{1}{6}$
- (b) $\frac{5}{6}$
- (c) $\frac{5}{6}$
- (d) 1

Previously asked in: 2026 30/1/1 Q18

◆ **Probability**

14.1 Probability — A Theoretical Approach

Q33. § 14.1 Probability — A Theoretical Approach**[4]**

A middle school decided to run the following spinner game as a fund-raiser on Christmas Carnival.

Making Purple : Spin each spinner once. Blue and red make purple. So, if one spinner shows Red (R) and the other shows Blue (B), you win. Such an outcome is denoted as a win.

Based on the above, answer the following questions :

- (i) List all possible outcomes of the game. [1]
- (ii) Find the probability of making purple. [1]
- (iii) For each win, a participant gets ₹10, but if he/she loses, he/she has to pay ₹5 to the school. If 99 participants played, calculate how much fund could the school have collected. OR If the same amount of ₹5 has been decided for winning or losing the game, then how much fund had been collected by school ? (Number of participants = 99) [2]

Previously asked in: 2023 30/5/1 Q37

♦ Probability

14.1 Probability — A Theoretical Approach

Expected Value / Fund Collection using Probability

Q34. § Chapter Opening and Epigraph § 14.2 Summary**[1]**

Which of the following numbers cannot be the probability of happening of an event ?

- (a) 0
- (b) 7
- (c) $0 \cdot 01$
- (d) $\frac{0 \cdot 07}{3}$

Previously asked in: 2023 30/5/1 Q17

♦ Probability

14.1 Probability — A Theoretical Approach

Q35. § 14.1 Probability — A Theoretical Approach**[1]**

Two coins are tossed together. The probability of getting at least one tail is :

- (a) $\frac{1}{4}$
- (b) $\frac{1}{2}$
- (c) $\frac{3}{4}$
- (d) 1

Previously asked in: 2023 30/5/1 Q15

♦ Probability

14.1 Probability — A Theoretical Approach

Q36. § 14.1 Probability — A Theoretical Approach**[2]**

If a fair coin is tossed twice, find the probability of getting 'atmost one head'.

Previously asked in: 2023 30/4/1 Q24

♦ Probability

14.1 Probability — A Theoretical Approach

Q37. § Chapter Opening and Epigraph § 14.2 Summary**[1]**

Assertion (A) : The probability that a leap year has 53 Sundays is $\frac{2}{7}$.

Reason (R) : The probability that a non-leap year has 53 Sundays is $\frac{5}{7}$.

Choose the correct option :

- (a) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
- (b) Both Assertion (A) and Reason (R) are true and Reason (R) is not the correct explanation of Assertion (A).
- (c) Assertion (A) is true but Reason (R) is false.
- (d) Assertion (A) is false but Reason (R) is true.

Previously asked in: 2023 30/4/1 Q19

◆ Probability

14.1 Probability — A Theoretical Approach

Probability related to days in a leap/non-leap year

Q38. § Chapter Opening and Epigraph § 14.1 Probability — A Theoretical Approach § 14.2 Summary**[1]**

A card is drawn at random from a well-shuffled pack of 52 cards. The probability that the card drawn is not an ace is :

- (a) $\frac{1}{13}$
- (b) $\frac{9}{13}$
- (c) $\frac{4}{13}$
- (d) $\frac{12}{13}$

Previously asked in: 2023 30/4/1 Q18

◆ Probability

14.1 Probability — A Theoretical Approach

Q39. § 14.1 Probability — A Theoretical Approach**[1]**

Two dice are thrown together. The probability of getting the difference of numbers on their upper faces equals to 3 is :

- (a) $\frac{1}{9}$
- (b) $\frac{2}{9}$
- (c) $\frac{1}{6}$
- (d) $\frac{1}{12}$

Previously asked in: 2023 30/4/1 Q11

◆ Probability

14.1 Probability — A Theoretical Approach

Q40. § Chapter Opening and Epigraph § 14.2 Summary**[4]**

Computer-based learning (CBL) refers to any teaching methodology that makes use of computers for information transmission. At an elementary school level, computer applications can be used to display multimedia lesson plans. A survey was done on 1000 elementary and secondary schools of Assam and they were classified by the number of computers they had.

One school is chosen at random. Then:

- (i) Find the probability that the school chosen at random has more than 100 computers. [1]
- (ii) Find the probability that the school chosen at random has 50 or fewer computers. [2]
- (iii) Find the probability that the school chosen at random has 10 or less than 10 computers. [1]

Previously asked in: 2023 30/2/1 Q38

♦ Probability

14.1 Probability — A Theoretical Approach

Q41. § 14.1 Probability — A Theoretical Approach**[1]**

If three coins are tossed simultaneously, what is the probability of getting at most one tail?

- (a) $\frac{3}{8}$
- (b) $\frac{4}{8}$
- (c) $\frac{5}{8}$
- (d) $\frac{7}{8}$

Previously asked in: 2023 30/2/1 Q11

♦ Probability

14.1 Probability — A Theoretical Approach

Q42. § Chapter Opening and Epigraph**[1]**

A bag contains 100 cards numbered 1 to 100. A card is drawn at random from the bag. What is the probability that the number on the card is a perfect cube?

- (a) $\frac{1}{20}$
- (b) $\frac{3}{50}$
- (c) $\frac{1}{25}$
- (d) $\frac{7}{100}$

Previously asked in: 2023 30/2/1 Q7

♦ Probability

14.1 Probability — A Theoretical Approach

Q43. § 14.1 Probability — A Theoretical Approach**[3]**

Three unbiased coins are tossed simultaneously. Find the probability of getting :

- (i) at least one head. [1]
- (ii) exactly one tail. [1]
- (iii) two heads and one tail. [1]

Previously asked in: 2024 30/5/1 Q30

♦ Probability

14.1 Probability — A Theoretical Approach

Q44. § Chapter Opening and Epigraph**[2]**

A carton consists of 60 shirts of which 48 are good, 8 have major defects and 4 have minor defects. Nigam, a trader, will accept the shirts which are good but Anmol, another trader, will only reject the shirts which have major defects. One shirt is drawn at random from the carton. Find the probability that it is acceptable to Anmol.

Previously asked in: 2024 30/5/1 Q25

♦ Probability

14.1 Probability — A Theoretical Approach

Q45. § Chapter Opening and Epigraph § 14.1 Probability – A Theoretical Approach**[1]**

One ticket is drawn at random from a bag containing tickets numbered 1 to 40. The probability that the selected ticket has a number which is a multiple of 7 is :

- A $\frac{1}{7}$
- B $\frac{1}{8}$
- C $\frac{1}{5}$
- D $\frac{7}{40}$

Previously asked in: 2024 30/5/1 Q7

♦ Probability

14.1 Probability – A Theoretical Approach

Q46. § Chapter Opening and Epigraph § 14.1 Probability – A Theoretical Approach**[3]**

A box contains 90 discs which are numbered 1 to 90. If one disc is drawn at random from the box, find the probability that it bears a :

- (i) 2-digit number less than 40. [1]
- (ii) number divisible by 5 and greater than 50. [1]
- (iii) a perfect square number. [1]

Previously asked in: 2024 30/4/1 Q27(b) (OR-2)

♦ Probability

14.1 Probability – A Theoretical Approach

Q47. § 14.1 Probability – A Theoretical Approach**[3]**

Three coins are tossed simultaneously. What is the probability of getting

- (i) at least one head ? [1]
- (ii) exactly two tails ? [1]
- (iii) at most one tail ? [1]

Previously asked in: 2024 30/4/1 Q27(a) (OR-1)

♦ Probability

14.1 Probability – A Theoretical Approach

Q48. § Chapter Opening and Epigraph § 14.2 Summary**[1]**

For an event E , if $P(E) + P(\bar{E}) = q$, then the value of $q^2 - 4$ is :

- A -3
- B 3
- C 5
- D -5

Previously asked in: 2024 30/4/1 Q17

♦ Probability

14.1 Probability – A Theoretical Approach

Q49. § 14.1 Probability — A Theoretical Approach**[1]**

Two dice are tossed simultaneously. The probability of getting odd numbers on both the dice is :

- A $\frac{6}{36}$
B $\frac{36}{12}$
C $\frac{36}{9}$
D $\frac{36}{36}$

Previously asked in: 2024 30/4/1 Q8

♦ Probability

14.1 Probability — A Theoretical Approach

Q50. § Chapter Opening and Epigraph § 14.1 Probability — A Theoretical Approach**[2]**

One card is drawn at random from a well shuffled deck of 52 cards. Find the probability that the card drawn

(i) is queen of hearts; [1]

(ii) is not a jack. [1]

Previously asked in: 2024 30/3/1 Q21

♦ Probability

14.1 Probability — A Theoretical Approach

Q51. § Chapter Opening and Epigraph § 14.1 Probability — A Theoretical Approach § 14.2 Summary**[1]**

If a digit is chosen at random from the digits 1, 2, 3, 4, 5, 6, 7, 8, 9; then the probability that this digit is an odd prime number is :

- A $\frac{1}{3}$
B $\frac{3}{2}$
C $\frac{3}{4}$
D $\frac{9}{5}$

Previously asked in: 2024 30/3/1 Q15

♦ Probability

14.1 Probability — A Theoretical Approach

Q52. § 14.1 Probability — A Theoretical Approach § 14.2 Summary**[1]**

The probability of guessing the correct answer to a certain test question is $\frac{x}{6}$. If the probability of not guessing the correct answer to this question is $\frac{2}{3}$, then the value of x is :

- A 2
B 3
C 4
D 6

Previously asked in: 2024 30/3/1 Q4

♦ Probability

14.1 Probability — A Theoretical Approach

Q53. § 14.1 Probability — A Theoretical Approach

[1]

Two dice are thrown together. The probability that they show different numbers is :

- A $\frac{1}{6}$
 B $\frac{5}{6}$
 C $\frac{1}{3}$
 D $\frac{3}{2}$

Previously asked in: 2024 30/3/1 Q3

◆ Probability

14.1 Probability — A Theoretical Approach

Q54. § Chapter Opening and Epigraph

[4]

In a survey on holidays, 120 people were asked to state which type of transport they used on their last holiday. The following pie chart shows the results of the survey.

In a survey on holidays, 120 people were asked to state which type of transport they used on their last holiday. The following pie chart shows the results of the survey. Observe the pie chart and answer the following questions:

- (i) If one person is selected at random, find the probability that he/she travelled by bus or ship. [1]
 (ii) Which is most favourite mode of transport and how many people used it? [1]
 (iii) A person is selected at random. If the probability that he did not use train is $\frac{4}{5}$, find the number of people who used train. OR The probability that randomly selected person used aeroplane is $\frac{7}{60}$. Find the revenue collected by air company at the rate of ₹5,000 per person. [2]

Previously asked in: 2024 30/2/1 Q38

◆ Probability

14.1 Probability — A Theoretical Approach

Q55. § Chapter Opening and Epigraph

[1]

A box contains cards numbered 6 to 55. A card is drawn at random from the box. The probability that the drawn card has a number which is a perfect square, is

- (A) $\frac{7}{50}$
 (B) $\frac{7}{55}$
 (C) $\frac{1}{10}$
 (D) $\frac{5}{49}$

Previously asked in: 2024 30/2/1 Q13

◆ Probability

14.1 Probability — A Theoretical Approach

Q56. § Chapter Opening and Epigraph § 14.2 Summary

[1]

Two dice are rolled together. The probability of getting the sum of the two numbers to be more than 10, is

- (A) $\frac{1}{9}$
 (B) $\frac{1}{6}$
 (C) $\frac{7}{12}$
 (D) $\frac{1}{12}$

Previously asked in: 2024 30/2/1 Q8

◆ Probability

14.1 Probability — A Theoretical Approach

Q57. § 14.1 Probability — A Theoretical Approach**[3]**

Three unbiased coins are tossed simultaneously. Find the probability of getting:

- (a) exactly two tails [1]
- (b) at least one head [1]
- (c) at most two heads [1]

Previously asked in: 2025 30/3/1 Q30

◆ **Probability** 14.1 Probability — A Theoretical Approach

Q58. § Chapter Opening and Epigraph § 14.1 Probability — A Theoretical Approach § 14.2 Summary**[2]**

The probability of guessing the correct answer of a certain test question is $\frac{x}{12}$. If the probability of not guessing the correct answer is $\frac{2}{3}$, then find the value of x .

Previously asked in: 2025 30/3/1 Q22

◆ **Probability** 14.1 Probability — A Theoretical Approach

Q59. § 14.1 Probability — A Theoretical Approach**[1]**

A die is thrown once. The probability of getting a number which is not a factor of 36, is:

- A $\frac{1}{6}$
- B $\frac{1}{3}$
- C $\frac{1}{2}$
- D $\frac{5}{6}$

Previously asked in: 2025 30/3/1 Q14

◆ **Probability** 14.1 Probability — A Theoretical Approach

Q60. § Chapter Opening and Epigraph § 14.1 Probability — A Theoretical Approach § 14.2 Summary**[4]**

Rahul is a lucky charm for his cricket team. He has a jar of cards with numbers from 10 to 74. Before each match, he draws a card from the jar. If the card bears an even number, the team wins. If the number is even and divisible by 5, they win by a big margin. If the number is an odd number less than 30, they win by a small margin. And if the number is a prime number between 50 and 74, they lose.

Answer the following questions if Rahul draws a card today :

- (i) What is the probability that Rahul draws a card with an even number? [1]
- (ii) What is the probability that Rahul draws a card with an odd number less than 30? [1]
- (iii) What is the probability that Rahul draws a card with a prime number between 50 and 74? [2]

Previously asked in: 2025 30/2/1 Q37

◆ **Probability** 14.1 Probability — A Theoretical Approach

Q61. § Chapter Opening and Epigraph § 14.1 Probability — A Theoretical Approach**[1]**

In a cricket match, a batsman hits the boundary 7 times out of the 42 balls he plays. The probability of his not hitting a boundary is :

- A $\frac{1}{6}$
B $\frac{5}{6}$
C $\frac{7}{7}$
D $\frac{7}{42}$

Previously asked in: 2025 30/2/1 Q11

◆ Probability

14.1 Probability — A Theoretical Approach

Q62. § Chapter Opening and Epigraph § 14.2 Summary**[1]**

The probability of drawing an even prime number out of numbers from 1 to 30 is :

- A $\frac{1}{30}$
B $\frac{1}{15}$
C $\frac{1}{10}$
D 0

Previously asked in: 2025 30/2/1 Q2

◆ Probability

14.1 Probability — A Theoretical Approach

Q63. § 14.1 Probability — A Theoretical Approach**[3]**

Two dice are thrown at the same time. Determine the probability that the difference of the numbers on the two dice is 2.

Previously asked in: 2025 30/1/1 Q31

◆ Probability

14.1 Probability — A Theoretical Approach

Q64. § 14.2 Summary**[1]**

Assertion (A): The probability of selecting a number at random from the numbers 1 to 20 is 1.

Reason (R): For any event E , if $P(E) = 1$, then E is called a sure event.

Select the correct answer from the codes (A), (B), (C) and (D) given below.

- A Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).
B Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).
C Assertion (A) is true, but Reason (R) is false.
D Assertion (A) is false, but Reason (R) is true.

Previously asked in: 2025 30/1/1 Q19

◆ Probability

14.1 Probability — A Theoretical Approach

Q65. § Chapter Opening and Epigraph § 14.1 Probability – A Theoretical Approach**[1]**

A card is selected at random from a deck of 52 playing cards. The probability of it being a red face card is:

- A $\frac{3}{26}$
 B $\frac{3}{13}$
 C $\frac{1}{13}$
 D $\frac{1}{2}$

Previously asked in: 2025 30/1/1 Q13

◆ Probability

14.1 Probability – A Theoretical Approach

Q66.**[3]**

A bag contains 30 balls out of which ' m ' number of balls are blue in colour.

- (i) Find the probability that a ball drawn at random from the bag is not blue. [1]
 (ii) If 6 more blue balls are added in the bag, then the probability of drawing a blue ball will be $\frac{5}{4}$ times the probability of drawing a blue ball in the first case. Find the value of m . [2]

Previously asked in: 2026 30/4/1 Q30

◆ Probability

14.1 Probability – A Theoretical Approach

Q67. § 14.2 Summary**[1]**

Assertion (A): If probability of happening of an event is $0.2p$, $p > 0$, then p can't be more than 5.

Reason (R): $P(\bar{E}) = 1 - P(E)$ for an event E .

Select the correct answer from the options (A), (B), (C) and (D) as given below:

- A Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).
 B Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of Assertion (A).
 C Assertion (A) is true, but Reason (R) is false.
 D Assertion (A) is false, but Reason (R) is true.

Previously asked in: 2026 30/4/1 Q19

◆ Probability

14.1 Probability – A Theoretical Approach

Q68. § Chapter Opening and Epigraph § 14.1 Probability – A Theoretical Approach**[1]**

Two different dice are rolled together. The probability that both the obtained numbers are less than 4, is

- A $\frac{2}{9}$
 B $\frac{3}{36}$
 C $\frac{1}{4}$
 D $\frac{2}{3}$

Previously asked in: 2026 30/4/1 Q10

◆ Probability

14.1 Probability – A Theoretical Approach

Q69. § 14.2 Summary**[1]**

Assertion (A): In a cricket match, a batsman hits a boundary 9 times out of 45 balls he plays. The probability that in a given ball, he does not hit the boundary is $\frac{4}{5}$.

Reason (R): $P(E) + P(\text{not } E) = 1$

A Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

B Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).

C Assertion (A) is true, but Reason (R) is false.

D Assertion (A) is false, but Reason (R) is true.

Previously asked in: 2024 30/3/1 Q20

♦ Probability

14.1 Probability — A Theoretical Approach

Complementary Events: $P(E) + P(\text{not } E) = 1$

CBSE CLASS X

Mathematics Standard (041)

ANSWER KEY

From previous CBSE Board Exam questions

Code: CYDFDR	Questions: 69	Maximum Marks: 116	Generated: 2026-06-15 13:05
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Q1. § 13.4 Median of Grouped Data

[4]

BINGO is game of chance. The host has 75 balls numbered 1 through 75. Each player has a BINGO card with some numbers written on it. The participant cancels the number on the card when called out a number written on the ball selected at random. Whosoever cancels all the numbers on his/her card, says BINGO and wins the game. The table given below, shows the data of one such game where 48 balls were used before Tara said 'BINGO'.

Based on the above information, answer the following :

- (i) Write the median class. [1]
- (ii) When first ball was picked up, what was the probability of calling out an even number ? [1]
- (iii) Find median of the given data. [2]

Previously asked in: 2024 30/1/1 Q37

◆ Statistics

13.4 Median of Grouped Data

14.1 Probability — A Theoretical Approach

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Model Answer**(i) Median Class:**

Cumulative frequencies: 8, 17, 27, 39, 48.

$n = 48$, $n/2 = 24$. The cumulative frequency just exceeding 24 is 27 (class 30–45).

Median class = 30–45

(ii) Probability of calling an even number (first ball):

Even numbers from 1 to 75: 2, 4, 6, ... 74 → **37 even numbers**

Total balls = 75

$$P(\text{even}) = \frac{37}{75}$$

(iii) Median:

Median class = 30–45, $l = 30$, $f = 10$, $cf = 17$, $h = 15$, $n/2 = 24$

$$\begin{aligned} \text{Median} &= l + \frac{\frac{n}{2} - cf}{f} \times h = 30 + \frac{24 - 17}{10} \times 15 \\ &= 30 + \frac{7 \times 15}{10} = 30 + 10.5 = \mathbf{40.5} \end{aligned}$$

Source: Statistics, Median of Grouped Data

Explanation

- **Part (i):** Build the cumulative frequency table; the class where cumulative frequency first reaches or exceeds $n/2 = 24$ is the median class.
- **Part (ii):** Balls are 1–75; even numbers are 2, 4, ..., 74 = 37 numbers. Total = 75. Many students mistakenly write 38 — remember 75 is odd so evens = 37.
- **Part (iii):** Apply the standard median formula. Carry cf of the class *before* the median class (17, not 27). This is a common error to avoid.

Q2. § Chapter Opening and Epigraph**[2]**

In a pack of 52 playing cards one card is lost. From the remaining cards, a card is drawn at random. Find the probability that the drawn card is queen of heart, if the lost card is a black card.

Previously asked in: 2024 30/1/1 Q22

◆ **Probability**

14.1 Probability — A Theoretical Approach

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Model Answer

Since the lost card is a black card, the remaining cards = $52 - 1 = 51$.

There is only 1 queen of heart in the deck, and since the lost card is black (not the queen of heart), the queen of heart is still present among the remaining cards.

$$P(\text{queen of heart}) = \frac{1}{51}$$

Source: Chapter 14 – Probability, Section 14.1

Explanation

- Key point: the lost card is a **black** card (clubs or spades). The queen of heart is a **red** card, so it is definitely still in the remaining 51 cards → favourable outcomes = 1.
- Total outcomes = 51 (not 52), because one card is already lost.
- Many students mistakenly write 52 in the denominator — always reduce by the number of lost/removed cards.

Q3. § Chapter Opening and Epigraph § 14.1 Probability — A Theoretical Approach § 14.2 Summary**[1]**

Two dice are rolled together. The probability of getting sum of numbers on the two dice as 2, 3 or 5, is :

- (a) $\frac{7}{36}$
 (b) $\frac{11}{36}$
 (c) $\frac{5}{36}$
 (d) $\frac{4}{9}$

Previously asked in: 2024 30/1/1 Q16

◆ **Probability**

14.1 Probability — A Theoretical Approach

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Model Answer

(a) $\frac{7}{36}$

Total outcomes = 36. Favourable outcomes (sum = 2, 3, or 5): sum 2 → (1,1); sum 3 → (1,2),(2,1); sum 5 → (1,4), (2,3),(3,2),(4,1) → 7 outcomes. $P = \frac{7}{36}$.

Explanation

Count outcomes for each sum separately: sum of 2 gives 1 outcome, sum of 3 gives 2 outcomes, sum of 5 gives 4 outcomes — total 7. Divide by 36 (total outcomes when two dice are rolled). Students often miss that sum = 4 is not asked, so don't include those pairs.

Q4. § Chapter Opening and Epigraph § 14.1 Probability — A Theoretical Approach § 14.2 Summary

[1]

From the data 1, 4, 7, 9, 16, 21, 25, if all the even numbers are removed, then the probability of getting at random a prime number from the remaining is :

- (a) $\frac{2}{5}$
- (b) $\frac{1}{5}$
- (c) $\frac{1}{7}$
- (d) $\frac{2}{7}$

Previously asked in: 2024 30/1/1 Q8

◆ **Probability** 14.1 Probability — A Theoretical Approach

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Model Answer

After removing even numbers (4, 16), the remaining data is: 1, 7, 9, 21, 25 → 5 numbers.

Prime numbers among these: 7 only → 1 prime.

$$P(\text{prime}) = \frac{1}{5}$$

Answer: (b) $\frac{1}{5}$

Explanation

- Even numbers in the set: 4 and 16. Remove them; remaining = {1, 7, 9, 21, 25} — 5 elements.
- Check primes: 1 is not prime; 7 is prime; 9 = 3×3 (not prime); 21 = 3×7 (not prime); 25 = 5×5 (not prime).
- Only 1 favourable outcome (7) out of 5 → P = 1/5.
- Common mistake: treating 1 as prime or 9/21/25 as prime — avoid this.

Q5. § Chapter Opening and Epigraph

[1]

If the probability of a player winning a game is 0.79, then the probability of his losing the same game is :

- (a) 1.79
- (b) 0.31
- (c) 0.21%
- (d) 0.21

Previously asked in: 2024 30/1/1 Q2

◆ **Probability** 14.1 Probability — A Theoretical Approach

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Model Answer

(d) 0.21

$$P(\text{losing}) = 1 - P(\text{winning}) = 1 - 0.79 = \mathbf{0.21}$$

Source: Chapter 14, Section 14.1 (Complementary events: $P(\bar{E}) = 1 - P(E)$)

Explanation

Using the complementary event rule: $P(E) + P(\text{not } E) = 1$. Since $P(\text{winning}) = 0.79$, $P(\text{losing}) = 1 - 0.79 = 0.21$.

Options (a) and (b) are incorrect calculations; (c) is wrong as $0.21\% \neq 0.21$.

Q6. § 14.2 Summary**[2]**

A bag contains 25 balls. Some of them are yellow and others are green. One ball is drawn at random. If probability of getting a green ball is $\frac{3}{5}$, then find the number of yellow balls.

Previously asked in: 2026 30/5/1 Q25

♦ Probability

14.1 Probability — A Theoretical Approach

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Model Answer

Total number of balls = 25

$P(\text{green ball}) = \frac{3}{5}$

Number of green balls = $(\frac{3}{5}) \times 25 = 15$

Using complementary events: $P(\text{yellow}) = 1 - \frac{3}{5} = \frac{2}{5}$

Number of yellow balls = $(\frac{2}{5}) \times 25 = \mathbf{10}$

Source: Chapter 14, Section 14.1

Explanation

The examiner expects you to first find the number of green balls (or directly use the complement), then subtract from 25. Both methods are acceptable, but showing the complementary probability step earns full marks. Key formula used: Number of favourable outcomes = $P(E) \times \text{Total outcomes}$.

Q7. § Chapter Opening and Epigraph § 14.2 Summary**[1]**

Meena calculates that the probability of her winning the first prize in a lottery is 0.08. If total 800 tickets were sold, the number of tickets bought by her, is

- (A) 64
- (B) 640
- (C) 100
- (D) 10

Previously asked in: 2026 30/5/1 Q16

♦ Probability

14.1 Probability — A Theoretical Approach

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Model Answer

(A) 64

Number of tickets bought by Meena = $P(\text{winning}) \times \text{Total tickets} = 0.08 \times 800 = \mathbf{64}$

Source: Chapter 14, Section 14.1

Explanation

Use the formula: $P(E) = (\text{Favourable outcomes})/(\text{Total outcomes})$. Here, favourable outcomes = tickets bought by Meena, total outcomes = 800. So, tickets = $0.08 \times 800 = 64$. Don't confuse 0.08 with 0.8 (which gives 640, a common mistake).

Q8. § Chapter Opening and Epigraph § 14.1 Probability — A Theoretical Approach

[1]

A card is drawn at random from a well shuffled deck of 52 playing cards. The probability that it is either a ten or a king is

- (A) $\frac{1}{26}$
 (B) $\frac{2}{13}$
 (C) $\frac{1}{13}$
 (D) $\frac{8}{26}$

Previously asked in: 2026 30/5/1 Q11

♦ Probability 14.1 Probability — A Theoretical Approach

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Model Answer

(B) $\frac{2}{13}$

There are 4 tens and 4 kings in a deck, giving 8 favourable outcomes.

$$P(\text{ten or king}) = \frac{8}{52} = \frac{2}{13}$$

Source: Chapter 14, Section 14.1

Explanation

A standard deck has 13 cards per suit $\times$ 4 suits = 52 cards. There are exactly 4 tens and 4 kings (one of each suit), so 8 favourable outcomes in total. Simplify $8/52$ by dividing by 4 to get $2/13$. Option (D) shows $8/26$ which is not fully simplified and is not a standard answer choice.

Q9. § 14.2 Summary

[1]

Which of the following can not be the probability of an event?

- (A) $\frac{39}{100}$
 (B) $\frac{0.001}{20}$
 (C) $\frac{10}{0.2}$
 (D) 10%

Previously asked in: 2026 30/5/1 Q10

♦ Probability 14.1 Probability — A Theoretical Approach

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Model Answer

(C) $\frac{10}{0.2}$

Since $\frac{10}{0.2} = 50$, which is greater than 1, it cannot be the probability of an event (probability must satisfy $0 \leq P(E) \leq 1$).

Source: Chapter 14, Section 14.1

Explanation

The key rule is $0 \leq P(E) \leq 1$. Check each option: (A) $39/100 = 0.39$ ✓, (B) $0.001/20 = 0.00005$ ✓, (D) $10\% = 0.1$ ✓ — all lie between 0 and 1. Only (C) equals 50, which exceeds 1, so it is impossible as a probability.

Q10. § Chapter Opening and Epigraph § 14.1 Probability — A Theoretical Approach

[4]

"Eight Ball" is a game played on a pool table with 15 balls numbered 1 to 15 and a "cue ball" that is solid and white. Of the 15 numbered balls, eight are solid (non-white) coloured and numbered 1 to 8 and seven are striped balls numbered 9 to 15.

The 15 numbered pool balls (no cue ball) are placed in a large bowl and mixed, then one ball is drawn out at random.

Based on the above information, answer the following questions:

- (i) What is the probability that the drawn ball bears number 8? [1]
- (ii) What is the probability that the drawn ball bears an even number? [2]
- (iii) What is the probability that the drawn ball is a solid coloured and bears an even number? [1]

Previously asked in: 2023 30/6/1 Q38

◆ Probability 14.1 Probability — A Theoretical Approach

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Model Answer

Total number of balls = 15

(i) Only one ball bears number 8.

$$P(\text{ball numbered 8}) = \frac{1}{15}$$

(ii) Even-numbered balls from 1 to 15: 2, 4, 6, 8, 10, 12, 14 → 7 balls

$$P(\text{even number}) = \frac{7}{15}$$

(iii) Solid coloured balls are numbered 1 to 8. Even numbers among these: 2, 4, 6, 8 → 4 balls

$$P(\text{solid and even}) = \frac{4}{15}$$

Source: Probability (Case Study based on pool ball problem)

Explanation

- **Total outcomes = 15** (only numbered balls, cue ball excluded).
- For (ii), list all even numbers up to 15 carefully — students often miss that 7 even numbers exist (2,4,6,8,10,12,14).
- For (iii), restrict to solid balls (1–8) first, then pick even ones (2,4,6,8) = 4. Do not include striped even numbers (10,12,14) here.
- Write the favourable outcomes explicitly before writing the fraction — examiners award step marks.

Q11. § Chapter Opening and Epigraph § 14.1 Probability — A Theoretical Approach

[1]

A card is drawn at random from a well shuffled deck of 52 playing cards. The probability of getting a face card is

- A $\frac{1}{2}$
- B $\frac{3}{13}$
- C $\frac{4}{13}$
- D $\frac{1}{13}$

Previously asked in: 2023 30/6/1 Q15

◆ Probability 14.1 Probability — A Theoretical Approach

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Model Answer

Option B: $\frac{3}{13}$

In a deck of 52 cards, face cards are kings, queens and jacks — 4 of each suit, so 12 face cards total.

$$P(\text{face card}) = \frac{12}{52} = \frac{3}{13}$$

Source: Chapter 14, Section 14.1

Explanation

The key fact from the textbook: "Kings, queens and jacks are called face cards." There are 4 suits $\times$ 3 face cards = 12 face cards out of 52 total cards. A common mistake is counting aces as face cards — they are **not**. Simplify 12/52 by dividing both by 4 to get 3/13.

Q12. § Chapter Opening and Epigraph

[1]

A bag contains 5 pink, 8 blue and 7 yellow balls. One ball is drawn at random from the bag. What is the probability of getting neither a blue nor a pink ball?

- A $\frac{1}{4}$
- B $\frac{2}{5}$
- C $\frac{7}{20}$
- D $\frac{13}{20}$

Previously asked in: 2023 30/6/1 Q11

◆ Probability 14.1 Probability — A Theoretical Approach

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Model Answer

Total balls = 5 + 8 + 7 = 20. Neither blue nor pink means only yellow balls are favourable. Number of yellow balls = 7.

$$P(\text{neither blue nor pink}) = \frac{7}{20}$$

Answer: C $\frac{7}{20}$

Explanation

"Neither blue nor pink" means the ball must be yellow. Count yellow balls (7) over total balls (20). Eliminate the other two colours — don't add them to the favourable count.

Q13. § Chapter Opening and Epigraph

[2]

A bag contains 4 red, 3 blue and 2 yellow balls. One ball is drawn at random from the bag. Find the probability that drawn ball is (i) red (ii) yellow.

Previously asked in: 2023 30/1/1 Q25

◆ Probability

14.1 Probability — A Theoretical Approach

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Model Answer

Total number of balls = $4 + 3 + 2 = 9$

(i) P(red ball)

Number of red balls = 4

$$P(\text{red}) = \frac{4}{9}$$

(ii) P(yellow ball)

Number of yellow balls = 2

$$P(\text{yellow}) = \frac{2}{9}$$

Source: Chapter 14, Section 14.1

Explanation

- Always find the **total** number of outcomes first (here, $4 + 3 + 2 = 9$).
- Apply the formula: $P(E) = (\text{Favourable outcomes}) \div (\text{Total outcomes})$.
- Each sub-part carries 1 mark; one line with the fraction is sufficient. Simplify if possible ($2/9$ is already in simplest form).

Q14. § Chapter Opening and Epigraph § 14.1 Probability — A Theoretical Approach § 14.2 Summary

[1]

In a group of 20 people, 5 can't swim. If one person is selected at random, then the probability that he/she can swim, is

- A $\frac{3}{4}$
 B $\frac{1}{3}$
 C 1
 D $\frac{1}{4}$

Previously asked in: 2023 30/1/1 Q18

◆ Probability

14.1 Probability — A Theoretical Approach

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Model Answer

Total people = 20; people who **can** swim = $20 - 5 = 15$.

$$P(\text{can swim}) = \frac{15}{20} = \frac{3}{4}$$

Answer: A $\frac{3}{4}$

Source: Chapter 14, Section 14.1

Explanation

Subtract the 5 non-swimmers from 20 to get 15 favourable outcomes, then apply the formula $P(E) = \text{favourable outcomes} \div \text{total outcomes}$. Examiner expects the subtraction step shown and the fraction simplified to $3/4$.

Q15. § Chapter Opening and Epigraph § 14.2 Summary

[1]

A girl calculates that the probability of her winning the first prize in a lottery is 0.08. If 6000 tickets are sold, how many tickets has she bought ?

- A 40
 B 240
 C 480
 D 750

Previously asked in: 2023 30/1/1 Q17

◆ Probability

14.1 Probability — A Theoretical Approach

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Model Answer

Option C: 480

Number of tickets bought = $P(\text{winning}) \times \text{Total tickets} = 0.08 \times 6000 = 480$

Source: Chapter 14, Probability — A Theoretical Approach

Explanation

Use the formula: $P(E) = (\text{Favourable outcomes})/(\text{Total outcomes})$. Here, favourable outcomes = tickets she bought, total outcomes = 6000. So, tickets bought = $0.08 \times 6000 = 480$. Watch out for option B (240) — a common error from using 0.04 instead of 0.08.

Q16. § Chapter Opening and Epigraph § 14.2 Summary**[1]**

Probability of happening of an event is denoted by p and probability of non-happening of the event is denoted by q .
Relation between p and q is

A $p + q = 1$

B $p = 1, q = 1$

C $p = q - 1$

D $p + q + 1 = 0$

Previously asked in: 2023 30/1/1 Q16

♦ Probability

14.1 Probability — A Theoretical Approach

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Model Answer

Option A: $p + q = 1$

Since $p = P(E)$ and $q = P(\text{not } E)$, and $P(E) + P(\text{not } E) = 1$, the relation between p and q is **$p + q = 1$** .

Source: Chapter 14, Section 14.2 (Summary, Point 6)

Explanation

The textbook states: "For any event E , $P(E) + P(\bar{E}) = 1$, where $\bar{E}$ stands for 'not E '." Here $p = P(E)$ and $q = P(\bar{E})$, so $p + q = 1$. This is the complementary events rule — the most fundamental relation in probability. Examiners expect you to directly identify option A without calculation.

Q17. § Chapter Opening and Epigraph**[2]**

The number of red balls in a bag is three more than the number of black balls. If the probability of drawing a red ball at random from the given bag is $\frac{12}{23}$, find the total number of balls in the given bag.

Previously asked in: 2025 30/6/1 Q25

◆ **Probability**

14.1 Probability — A Theoretical Approach

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Model Answer

Let the number of black balls = x

Then, number of red balls = $x + 3$

Total balls = $x + (x + 3) = 2x + 3$

Given: $P(\text{red ball}) = \frac{12}{23}$

$$\frac{x + 3}{2x + 3} = \frac{12}{23}$$

$$23(x + 3) = 12(2x + 3)$$

$$23x + 69 = 24x + 36$$

$$x = 33$$

Total number of balls = $2(33) + 3 = 69$

Source: Chapter 14, Section 14.1

Explanation

- Set up a variable for black balls; red balls = black + 3; total = their sum.
- Use the theoretical probability formula: favourable outcomes ÷ total outcomes = $12/23$.
- Cross-multiply and solve for x , then substitute back to find the total.
- Examiners award 1 mark for correct equation setup and 1 mark for the correct final answer (69).

Q18. § 14.2 Summary

[1]

Directions: A statement of Assertion (A) is followed by a statement of Reason (R). Choose the correct option.

In an experiment of throwing a die,

Assertion (A): Event E_1 : getting a number less than 3 and Event E_2 : getting a number greater than 3 are complementary events.

Reason (R): If two events E and F are complementary events, then $P(E) + P(F) = 1$.

- A Both Assertion (A) and Reason (R) are true and Reason (R) is correct explanation of Assertion (A).
- B Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of Assertion (A).
- C Assertion (A) is true, but Reason (R) is false.
- D Assertion (A) is false, but Reason (R) is true.

Previously asked in: 2025 30/6/1 Q20

◆ Probability 14.1 Probability — A Theoretical Approach

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Model Answer

Option (D): Assertion (A) is false, but Reason (R) is true.

E_1 (less than 3) = {1, 2}, E_2 (greater than 3) = {4, 5, 6}. These are not complementary as they don't cover all outcomes (3 is excluded). Reason (R) is correct: $P(E) + P(\bar{E}) = 1$.

Explanation

Complementary events must together cover the entire sample space with no outcome left out. Here, outcome {3} belongs to neither E_1 nor E_2 , so their union $\neq$ sample space, making A false. The Reason is a standard, correct definition from the textbook (Summary point 6), so R is true.

Q19. § 14.1 Probability — A Theoretical Approach

[1]

Two coins are tossed simultaneously. The probability of getting atleast one head is

- A $\frac{1}{4}$
- B $\frac{1}{2}$
- C $\frac{3}{4}$
- D 1

Previously asked in: 2025 30/6/1 Q8

◆ Probability 14.1 Probability — A Theoretical Approach

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Model Answer

Option C: $\frac{3}{4}$

When two coins are tossed, total outcomes = {HH, HT, TH, TT} = 4. Favourable outcomes (at least one head) = {HH, HT, TH} = 3. So, $P(\text{at least one head}) = \frac{3}{4}$.

Source: Chapter 14, Example 9

Explanation

"At least one head" means one or more heads — include all outcomes except TT. Students often wrongly count only 3 outcomes total (HH, HT/TH, TT), giving $\frac{1}{3}$; remember HT and TH are **different** outcomes.

Q20. § Chapter Opening and Epigraph § 14.1 Probability – A Theoretical Approach**[2]**

While shuffling a pack of 52 cards, one card was accidentally dropped. Find the probability that the dropped card

- (i) is not a face card.
- (ii) is a black king.

Previously asked in: 2025 30/5/1 Q25

♦ Probability 14.1 Probability – A Theoretical Approach

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Model Answer

Total cards = 52. Face cards (kings, queens, jacks) = 12.

(i) P(not a face card)

Number of non-face cards = $52 - 12 = 40$

$$P(\text{not a face card}) = \frac{40}{52} = \frac{10}{13}$$

(ii) P(a black king)

There are 2 black kings (king of spades and king of clubs).

$$P(\text{black king}) = \frac{2}{52} = \frac{1}{26}$$

Source: Chapter 14 – Probability, Section 14.1

Explanation

- A standard deck has 52 cards: 4 suits (spades, clubs – black; hearts, diamonds – red), 13 cards each.
- Face cards = kings + queens + jacks = $4 + 4 + 4 = 12$.
- Black kings = king of spades + king of clubs = **2**.
- Examiners expect the formula, substitution, and simplified fraction for each part. Always simplify the fraction fully.

Q21. § 14.1 Probability — A Theoretical Approach § 14.2 Summary

[1]

The probability of getting a composite number greater than 3 on throwing a die is

- A $\frac{1}{6}$
 B $\frac{1}{3}$
 C $\frac{1}{2}$
 D $\frac{2}{3}$

Previously asked in: 2025 30/5/1 Q18

◆ Probability

14.1 Probability — A Theoretical Approach

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Model Answer

Option B: $\frac{1}{3}$

On a die, possible outcomes = {1, 2, 3, 4, 5, 6}. Composite numbers greater than 3 are 4 and 6 (2 outcomes). $P = \frac{2}{6} = \frac{1}{3}$.

Explanation

Remember: composite numbers have more than two factors. Among 1–6, composites are 4 and 6 ($4 = 2 \times 2$, $6 = 2 \times 3$). The number 1 is neither prime nor composite; 2, 3, 5 are prime. Since the question asks for composites **greater than 3**, only 4 and 6 qualify — giving 2 favourable outcomes out of 6.

Q22. § Chapter Opening and Epigraph

[1]

A bag contains red balls and black balls in the ratio 3 : 7. A ball is drawn at random. The probability that ball so drawn is black in colour, is

- A $\frac{3}{7}$
 B 0.3
 C 0.7
 D $\frac{1}{7}$

Previously asked in: 2025 30/5/1 Q7

◆ Probability

14.1 Probability — A Theoretical Approach

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Model Answer

Option C: 0.7

Ratio of red : black = 3 : 7, so total parts = 10. $P(\text{black}) = 7/10 = \mathbf{0.7}$.

Source: Chapter 14, Section 14.1

Explanation

Total balls = 3 + 7 = 10 parts. Favourable outcomes (black) = 7. $P(\text{black}) = 7/10 = 0.7$. Note that 3/7 is the *ratio* of red to black, not a probability — a common mistake to avoid.

Q23.

[2]

Two friends Anil and Ashraf were born in the December month in the year 2010. Find the probability that :

- (i) they share same date of birth. [1]
- (ii) they have different dates of birth. [1]

Previously asked in: 2025 30/4/1 Q25

◆ Probability

14.1 Probability — A Theoretical Approach

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Model Answer

December has 31 days, so the total number of possible dates of birth = 31.

(i) Probability that they share the same date of birth:

Favourable outcomes = 31 (Anil can be born on any of 31 dates; Ashraf's birthday matches if born on that same date)

$$P(\text{same date}) = \frac{31}{31 \times 31} = \frac{1}{31}$$

(ii) Probability that they have different dates of birth:

$$P(\text{different dates}) = 1 - P(\text{same date}) = 1 - \frac{1}{31} = \frac{30}{31}$$

Source: Chapter 14, Section 14.1 — Probability: A Theoretical Approach

Explanation

- December has **31 days**, so each person can be born on any of 31 dates. Total equally likely outcomes for the pair = $31 \times 31 = 961$.
- Favourable outcomes for the same date = 31 (any matching pair like 1-1, 2-2, ... 31-31).
- For part (ii), use the complementary event rule: $P(\text{different}) = 1 - P(\text{same})$. This is the standard method as shown in Example 6 of the chapter.

Q24. § 14.1 Probability — A Theoretical Approach

[1]

A pair of dice is thrown. The probability that sum of numbers appearing on top faces is at most 10 is :

- (a) $\frac{1}{11}$
- (b) $\frac{10}{11}$
- (c) $\frac{5}{6}$
- (d) $\frac{11}{12}$

Previously asked in: 2025 30/4/1 Q18

◆ Probability

14.1 Probability — A Theoretical Approach

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Model Answer

(d) $\frac{11}{12}$

Total outcomes = 36. Outcomes with sum **more than 10** (i.e., sum 11 or 12): (5,6),(6,5),(6,6) = 3 outcomes. So

$$P(\text{sum} \leq 10) = \frac{36 - 3}{36} = \frac{33}{36} = \frac{11}{12}.$$

Source: Chapter 14, Section 14.1

Explanation

- "At most 10" means $\text{sum} \leq 10$, so use the complement: find cases where $\text{sum} > 10$ (sum = 11 or 12), subtract from 36.
- Sum 11: (5,6),(6,5) → 2 outcomes; Sum 12: (6,6) → 1 outcome. Total = 3.
- $P(\text{sum} \leq 10) = 1 - 3/36 = 33/36 = \mathbf{11/12}$. Option (d) is correct.

Q25. § Chapter Opening and Epigraph § 14.2 Summary

[1]

The number of red balls in a bag is 10 more than the number of black balls. If the probability of drawing a red ball at random from this bag is $\frac{3}{5}$, then the total number of balls in the bag is :

- (a) 50
- (b) 60
- (c) 80
- (d) 40

Previously asked in: 2025 30/4/1 Q5

◆ Probability

14.1 Probability — A Theoretical Approach

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Model Answer

(d) 25

Wait — let me solve: Let black balls = x, red balls = x + 10. $P(\text{red}) = (x+10)/(2x+10) = 3/5 \rightarrow 5(x+10) = 3(2x+10) \rightarrow 5x+50 = 6x+30 \rightarrow x = 20$. Total = $2(20)+10 = \mathbf{50}$.

Answer: (a) 50

Explanation

Set black balls = x, red balls = x + 10, total = 2x + 10. Use $P(\text{red}) = 3/5$ to form the equation and solve for x = 20, giving total = 50. Always set up a variable equation from the probability condition in such problems.

Q26. § Chapter Opening and Epigraph § 14.1 Probability — A Theoretical Approach § 14.2 Summary**[4]**

A group of friends wanted to play cards with two identical packs together. While shuffling the cards, three cards are dropped. Rest of the cards are shuffled and one card is drawn at random. Assuming that the dropped cards were a queen of hearts, a ten of spades and an ace of clubs, answer the following questions:

Assuming that the dropped cards were a queen of hearts, a ten of spades and an ace of clubs, answer the following questions:

- (i) Find the probability that the drawn card is a face card. [1]
- (ii) Find the probability that the drawn card is either a king or a queen. [1]
- (iii) Do you think that the probability of getting a queen was higher if none of the cards were dropped? Justify your answer. [2]

Previously asked in: 2026 30/3/1 Q37

◆ Probability 14.1 Probability — A Theoretical Approach Probability with combined/modified decks of cards

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Model Answer

Given: Two identical packs = 104 cards total. Cards dropped: queen of hearts, ten of spades, ace of clubs → Remaining cards = $104 - 3 = 101$ cards

(i) Probability of drawing a face card:

Two packs have $2 \times 12 = 24$ face cards. One queen is dropped, so face cards left = 23.

$$P(\text{face card}) = \frac{23}{101}$$

(ii) Probability of drawing a king or a queen:

Kings in two packs = 8; Queens in two packs = 8, but one queen dropped → queens left = 7.

Total kings or queens = $8 + 7 = 15$.

$$P(\text{king or queen}) = \frac{15}{101}$$

(iii) If no cards were dropped, total cards = 104 and queens = 8.

$$P(\text{queen}) = \frac{8}{104} = \frac{1}{13} \approx 0.077$$

With dropped cards, queens = 7 and total = 101.

$$P(\text{queen}) = \frac{7}{101} \approx 0.069$$

Since $\frac{1}{13} > \frac{7}{101}$, **yes**, the probability of getting a queen was higher when no cards were dropped.

Explanation

- Always start by calculating the new total ($104 - 3 = 101$) and updating counts for each suit/type after the drop.
- Face cards = Jack, Queen, King ($12 \text{ per pack} \times 2 = 24$ total).
- For part (iii), examiners expect both probabilities to be calculated and compared numerically — just saying "yes" without justification will lose a mark.

- Keep fractions unreduced unless simplification is straightforward, to avoid arithmetic errors.

Q27. § 14.1 Probability — A Theoretical Approach**[1]**

Two dice are rolled together. The probability that the sum of the numbers obtained is divisible by 6, is :

- A $\frac{1}{6}$
- B $\frac{11}{36}$
- C $\frac{1}{12}$
- D $\frac{1}{4}$

Previously asked in: 2026 30/3/1 Q5

♦ Probability

14.1 Probability — A Theoretical Approach

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Model Answer**Option (A)** $\frac{1}{6}$

Total outcomes = 36. Favourable outcomes (sum divisible by 6, i.e., sum = 6 or 12): (1,5),(2,4),(3,3),(4,2),(5,1), (6,6) → 6 outcomes. $P = 6/36 = 1/6$.

Explanation

Sums divisible by 6 from two dice can be 6 or 12 only (sum ranges from 2–12). List all pairs giving sum 6 (five pairs) and sum 12 (one pair) = 6 favourable outcomes out of 36. Students often miss (6,6) for sum 12 or miscount pairs for sum 6. Source: Chapter 14, Section 14.1

Q28. § 14.1 Probability — A Theoretical Approach § 14.2 Summary**[3]**

Two dice are thrown at the same time. Determine the probability that the (i) sum of the numbers on the two dice is 5, and (ii) difference of the numbers on the two dice is 3.

Previously asked in: 2026 30/2/1 Q31

♦ Probability 14.1 Probability — A Theoretical Approach

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Model Answer

Total possible outcomes when two dice are thrown = $6 \times 6 = 36$

(i) Sum of numbers on two dice = 5

Favourable outcomes: (1,4), (2,3), (3,2), (4,1) → **4 outcomes**

$$P(\text{sum} = 5) = \frac{4}{36} = \frac{1}{9}$$

(ii) Difference of numbers on two dice = 3

Favourable outcomes: (4,1), (5,2), (6,3), (1,4), (2,5), (3,6) → **6 outcomes**

$$P(\text{difference} = 3) = \frac{6}{36} = \frac{1}{6}$$

Source: Chapter 14, Example 13 / Exercise 14.1

Explanation

- Always state total outcomes = 36 first (1 mark).
- For "difference = 3," consider both $(a-b = 3)$ and $(b-a = 3)$, giving 6 pairs — students often miss half of these.
- List favourable outcomes explicitly; examiners award a method mark for correct listing even if the final fraction has an error.
- Simplify the fraction to its lowest terms.

Q29. § Chapter Opening and Epigraph § 14.2 Summary

[1]

The probability for a randomly selected number out of 1, 2, 3, 4, ..., 25 to be a composite number is :

- A $\frac{15}{25}$
- B $\frac{10}{25}$
- C $\frac{11}{25}$
- D $\frac{9}{25}$

Previously asked in: 2026 30/2/1 Q18

♦ Probability

14.1 Probability — A Theoretical Approach

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Model Answer

Composite numbers from 1 to 25: 4, 6, 8, 9, 10, 12, 14, 15, 16, 18, 20, 21, 22, 24, 25 → 15 composite numbers.

$$P(\text{composite}) = \frac{15}{25}$$

Answer: A $\frac{15}{25}$

Explanation

Composite numbers are numbers greater than 1 that are **not prime** (and not 1). Students often forget that 1 is neither prime nor composite, and may miscount primes (2,3,5,7,11,13,17,19,23 = 9 primes). So composites = 25 – 9 primes – 1 = **15**. Option A is correct.

Q30. § 14.1 Probability — A Theoretical Approach**[3]**

Two dice of different colours are thrown at the same time. Write down all the possible outcomes. What is the probability that :

- (i) same number appears on both the dice ?
- (ii) different number appears on both the dice ?

Previously asked in: 2026 30/1/1 Q31

♦ Probability

14.1 Probability — A Theoretical Approach

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Model Answer

When two dice are thrown, total number of possible outcomes = $6 \times 6 = 36$.

The possible outcomes are: (1,1), (1,2), ... (1,6), (2,1), (2,2), ... (2,6), (3,1) ... (6,6).

(i) Same number on both dice:

Favourable outcomes: (1,1), (2,2), (3,3), (4,4), (5,5), (6,6) → 6 outcomes

$$P(\text{same number}) = \frac{6}{36} = \frac{1}{6}$$

(ii) Different number on both dice:

$$P(\text{different number}) = 1 - P(\text{same number}) = 1 - \frac{1}{6} = \frac{5}{6}$$

Source: Chapter 14, Section 14.1

Explanation

- Always state total outcomes (36 for two dice) first — examiners check this.
- For part (i), list all 6 favourable outcomes explicitly to show working.
- For part (ii), use the complement rule: $P(\bar{E}) = 1 - P(E)$. This saves time and shows conceptual understanding, earning full marks without listing all 30 outcomes separately.
- Parts (i) and (ii) carry about 1 mark each; the sample space listing carries 1 mark.

Q31. § 14.2 Summary

[1]

Directions: Two statements are given, one labelled as Assertion (A) and the other labelled as Reason (R). Select the correct answer from the codes (a), (b), (c) and (d).

Assertion (A) : The probability that a leap year has 53 Mondays is $\frac{2}{7}$.

Reason (R) : The probability that a non-leap year has 53 Mondays is $\frac{5}{7}$.

- (a) Both, Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).
- (b) Both, Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).
- (c) Assertion (A) is true, but Reason (R) is false.
- (d) Assertion (A) is false, but Reason (R) is true.

Previously asked in: 2026 30/1/1 Q19

◆ Probability 14.1 Probability — A Theoretical Approach

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Model Answer

(c) Assertion (A) is true, but Reason (R) is false.

A leap year has 366 days = 52 weeks + 2 extra days. The 2 extra days can be any of 7 pairs; 2 pairs contain Monday, so $P(53 \text{ Mondays}) = \frac{2}{7} \checkmark$. A non-leap year has 365 days = 52 weeks + 1 extra day; only 1 out of 7 days can be Monday, so $P(53 \text{ Mondays}) = \frac{1}{7}$, not $\frac{5}{7}$. ✗

Source: Chapter 14, Probability

Explanation

- **Assertion** is correct: leap year has 2 odd days $\rightarrow$ 2 favourable pairs out of 7 $\rightarrow P = \frac{2}{7}$.
- **Reason** is wrong: non-leap year has only 1 odd day $\rightarrow$ only 1 day out of 7 can be Monday $\rightarrow P = \frac{1}{7}$, not $\frac{5}{7}$.
- So the answer is **(c)**. Remember: always verify both A and R independently before choosing the option.

Q32. § 14.1 Probability — A Theoretical Approach § 14.2 Summary

[1]

A die is thrown once. Probability of getting a number other than 3 is :

- (a) $\frac{1}{6}$
- (b) $\frac{3}{6}$
- (c) $\frac{5}{6}$
- (d) 1

Previously asked in: 2026 30/1/1 Q18

◆ Probability 14.1 Probability — A Theoretical Approach

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Model Answer

(c) $\frac{5}{6}$

Total outcomes = 6. Favourable outcomes (numbers other than 3) = {1, 2, 4, 5, 6} = 5. So $P = \frac{5}{6}$.

Explanation

The examiners expect you to identify the favourable outcomes (all numbers except 3) and apply the formula $P(E) = \text{favourable outcomes} \div \text{total outcomes}$. Alternatively, $P(\text{not } 3) = 1 - P(3) = 1 - \frac{1}{6} = \frac{5}{6}$ using complementary events.

Q33. § 14.1 Probability — A Theoretical Approach**[4]**

A middle school decided to run the following spinner game as a fund-raiser on Christmas Carnival.

Making Purple : Spin each spinner once. Blue and red make purple. So, if one spinner shows Red (R) and the other shows Blue (B), you win. Such an outcome is denoted as a win.

Based on the above, answer the following questions :

- (i) List all possible outcomes of the game. [1]
- (ii) Find the probability of making purple. [1]
- (iii) For each win, a participant gets ₹10, but if he/she loses, he/she has to pay ₹5 to the school. If 99 participants played, calculate how much fund could the school have collected. OR If the same amount of ₹5 has been decided for winning or losing the game, then how much fund had been collected by school ? (Number of participants = 99) [2]

Previously asked in: 2023 30/5/1 Q37

◆ **Probability**

14.1 Probability — A Theoretical Approach

Expected Value / Fund Collection using Probability

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Model Answer**(i) All possible outcomes:**

Assuming Spinner 1 has Red (R) and Blue (B), and Spinner 2 has Red (R) and Blue (B):

Possible outcomes = {(R, R), (R, B), (B, R), (B, B)}

(ii) Probability of making purple:

Winning outcomes (one R and one B) = (R, B) and (B, R) → 2 outcomes

$$P(\text{purple}) = \frac{2}{4} = \frac{1}{2}$$

(iii)

Out of 99 participants:

- Winners = $99 \times \frac{1}{2} \approx 50$ (approximately), Losers = 49
- School collects from losers: $49 \times ₹5 = ₹245$
- School pays winners: $50 \times ₹10 = ₹500$
- **Net fund collected** = ₹245 – ₹500 = –₹255 (school loses ₹255)

OR

If ₹5 for both win and loss:

- Winners = 50, Losers = 49
- School collects from losers: $49 \times ₹5 = ₹245$
- School pays winners: $50 \times ₹5 = ₹250$
- **Net fund** = ₹245 – ₹250 = –₹5 (school loses ₹5)

Explanation

- **Part (i):** Examiners expect all 4 sample space outcomes listed clearly.
- **Part (ii):** Two outcomes constitute a win, so probability = $2/4 = 1/2$. This is the key fraction to use in part (iii).
- **Part (iii):** With 99 participants and $P(\text{win}) = 1/2$, split as ~50 winners and 49 losers. Calculate inflow (from losers) minus outflow (to winners). In the main case the school actually *loses* money — write that clearly. The

OR part similarly yields a small net loss. Show all steps for full marks.

Q34. § Chapter Opening and Epigraph § 14.2 Summary

[1]

Which of the following numbers cannot be the probability of happening of an event ?

- (a) 0
- (b) 7
- (c) $0 \cdot 01$
- (d) $\frac{0 \cdot 07}{3}$

Previously asked in: 2023 30/5/1 Q17

◆ Probability

14.1 Probability — A Theoretical Approach

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Model Answer

(b) 7

Since probability of any event must satisfy $0 \leq P(E) \leq 1$, the value 7 exceeds 1 and hence cannot be a probability.

Source: Chapter 14, Section 14.2 Summary

Explanation

The key rule is $0 \leq P(E) \leq 1$. Check each option: 0 is valid (impossible event), 0.01 is between 0 and 1, and $\frac{0.07}{3} \approx 0.023$ is also between 0 and 1. Only 7 exceeds 1, making it impossible as a probability value. Examiners expect you to state the rule and identify the violating option.

Q35. § 14.1 Probability — A Theoretical Approach

[1]

Two coins are tossed together. The probability of getting at least one tail is :

- (a) $\frac{1}{4}$
- (b) $\frac{1}{2}$
- (c) $\frac{3}{4}$
- (d) 1

Previously asked in: 2023 30/5/1 Q15

◆ Probability

14.1 Probability — A Theoretical Approach

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Model Answer

(c) $\frac{3}{4}$

Sample space = {HH, HT, TH, TT}; total outcomes = 4. Outcomes with at least one tail = {HT, TH, TT} = 3. P(at least one tail) = $\frac{3}{4}$.

Explanation

"At least one tail" means one or more tails, so exclude only the all-heads outcome (HH). The complement method also works: P(no tail) = $\frac{1}{4}$, so P(at least one tail) = $1 - \frac{1}{4} = \frac{3}{4}$. This mirrors Example 9 of the textbook (at least one head $\rightarrow \frac{3}{4}$).

Q36. § 14.1 Probability — A Theoretical Approach**[2]**

If a fair coin is tossed twice, find the probability of getting 'atmost one head'.

Previously asked in: 2023 30/4/1 Q24

◆ Probability

14.1 Probability — A Theoretical Approach

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Model Answer

When a coin is tossed twice, the sample space is: {HH, HT, TH, TT} → Total outcomes = 4

Event: 'at most one head' means 0 or 1 head.

Favourable outcomes: {TT, HT, TH} → 3 outcomes

$$P(\text{at most one head}) = \frac{3}{4}$$

Source: Chapter 14 – Probability, Section 14.1

Explanation

- "At most one head" means **0 heads or 1 head** — do not include HH.
- Always list the sample space first; examiners award a mark for it.
- The 2 marks are typically split: 1 for correct sample space/favourable outcomes, 1 for the final probability.
- A common error is writing only 3 outcomes in the sample space (treating HT and TH as the same) — they are **distinct** outcomes.

Q37. § Chapter Opening and Epigraph § 14.2 Summary

[1]

Assertion (A) : The probability that a leap year has 53 Sundays is $\frac{2}{7}$.Reason (R) : The probability that a non-leap year has 53 Sundays is $\frac{5}{7}$.

Choose the correct option :

- (a) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
- (b) Both Assertion (A) and Reason (R) are true and Reason (R) is not the correct explanation of Assertion (A).
- (c) Assertion (A) is true but Reason (R) is false.
- (d) Assertion (A) is false but Reason (R) is true.

Previously asked in: 2023 30/4/1 Q19

◆ Probability

14.1 Probability — A Theoretical Approach

Probability related to days in a leap/non-leap year

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Model Answer**(c) Assertion (A) is true but Reason (R) is false.**

A leap year has 366 days = 52 weeks + 2 extra days. The 2 extra days can be any of 7 pairs; 2 pairs include Sunday, so $P(53 \text{ Sundays}) = \frac{2}{7}$. ✓

A non-leap year has 365 days = 52 weeks + 1 extra day. Only 1 of 7 days can be Sunday, so $P(53 \text{ Sundays}) = \frac{1}{7}$, not $\frac{5}{7}$. ✗

Source: Chapter 14, Probability — Theoretical Approach

Explanation

- **Assertion** is correct: leap year gives 2 extra days, and exactly 2 of the 7 possible pairs (Sat–Sun, Sun–Mon) contain a Sunday → $P = \frac{2}{7}$.
- **Reason** is wrong: a non-leap year gives only 1 extra day; $P(53 \text{ Sundays}) = \frac{1}{7}$, not $\frac{5}{7}$. ($\frac{5}{7}$ would be the probability of *not* getting a 53rd Sunday in a non-leap year... even that is $\frac{6}{7}$, so $\frac{5}{7}$ is simply incorrect.)
- Examiners expect you to justify *both* A and R briefly — don't just state the option.

Q38. § Chapter Opening and Epigraph § 14.1 Probability — A Theoretical Approach § 14.2 Summary**[1]**

A card is drawn at random from a well-shuffled pack of 52 cards. The probability that the card drawn is not an ace is :

- (a) $\frac{1}{13}$
(b) $\frac{13}{9}$
(c) $\frac{4}{13}$
(d) $\frac{12}{13}$

Previously asked in: 2023 30/4/1 Q18

♦ Probability 14.1 Probability — A Theoretical Approach

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Model Answer

(d) $\frac{12}{13}$

Total cards = 52; number of aces = 4; cards that are not aces = 48.

$$P(\text{not an ace}) = \frac{48}{52} = \frac{12}{13}$$

Source: Chapter 14, Example 4(ii)

Explanation

The examiner expects you to recall that a deck has 4 aces, leaving 48 non-ace cards. Dividing 48 by 52 and simplifying gives $\frac{12}{13}$. You can also use the complement: $1 - \frac{1}{13} = \frac{12}{13}$. Either method is acceptable; just show the key step.

Q39. § 14.1 Probability — A Theoretical Approach**[1]**

Two dice are thrown together. The probability of getting the difference of numbers on their upper faces equals to 3 is :

- (a) $\frac{1}{9}$
(b) $\frac{2}{9}$
(c) $\frac{1}{6}$
(d) $\frac{1}{12}$

Previously asked in: 2023 30/4/1 Q11

♦ Probability

14.1 Probability — A Theoretical Approach

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Model Answer

(a) $\frac{1}{9}$

Total outcomes = 36. Favourable outcomes (difference = 3): (1,4),(4,1),(2,5),(5,2),(3,6),(6,3) → 4 pairs. $P =$

$$\frac{4}{36} = \frac{1}{9}.$$

Explanation

List pairs where $|\text{die1} - \text{die2}| = 3$: (1,4),(4,1),(2,5),(5,2),(3,6),(6,3) — exactly 4 pairs out of 36. Examiners expect you to identify all favourable outcomes correctly; missing any pair is a common error.

Q40. § Chapter Opening and Epigraph § 14.2 Summary**[4]**

Computer-based learning (CBL) refers to any teaching methodology that makes use of computers for information transmission. At an elementary school level, computer applications can be used to display multimedia lesson plans. A survey was done on 1000 elementary and secondary schools of Assam and they were classified by the number of computers they had.

One school is chosen at random. Then:

- (i) Find the probability that the school chosen at random has more than 100 computers. [1]
- (ii) Find the probability that the school chosen at random has 50 or fewer computers. [2]
- (iii) Find the probability that the school chosen at random has 10 or less than 10 computers. [1]

Previously asked in: 2023 30/2/1 Q38

♦ Probability 14.1 Probability — A Theoretical Approach

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Model Answer

Total number of schools = 1000

(i) Schools with more than 100 computers = 80

$$P(\text{more than 100 computers}) = \frac{80}{1000} = \frac{2}{25}$$

(ii) Schools with 50 or fewer computers = schools in (1–10) + (11–20) + (21–50)
= 250 + 200 + 290 = 740

$$P(50 \text{ or fewer computers}) = \frac{740}{1000} = \frac{37}{50}$$

(iii) Schools with 10 or fewer computers = 250

$$P(10 \text{ or fewer computers}) = \frac{250}{1000} = \frac{1}{4}$$

Source: Statistics, Probability (Empirical/Experimental Probability)

Explanation

- Empirical probability = (Number of favourable outcomes) ÷ (Total outcomes). Here total = 1000.
- For part (ii), "50 or fewer" includes all three categories: 1–10, 11–20, and 21–50. Students often miss the 21–50 group — be careful.
- Always simplify the fraction to lowest terms; examiners check this.

Q41. § 14.1 Probability — A Theoretical Approach**[1]**

If three coins are tossed simultaneously, what is the probability of getting at most one tail?

- (a) $\frac{3}{8}$
 (b) $\frac{4}{8}$
 (c) $\frac{5}{8}$
 (d) $\frac{7}{8}$

Previously asked in: 2023 30/2/1 Q11

◆ Probability

14.1 Probability — A Theoretical Approach

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Model Answer

(c) $\frac{5}{8}$

Sample space = {HHH, HHT, HTH, THH, HTT, THT, TTH, TTT} → 8 outcomes. At most one tail: {HHH, HHT, HTH, THH} → 4 outcomes. $P = \frac{4}{8} = \frac{1}{2}$.

Source: Chapter 14, Section 14.1

Explanation

"At most one tail" means 0 tails or exactly 1 tail. List all 8 equally likely outcomes for three coins; count those with 0 or 1 tail (that's $1 + 3 = 4$ outcomes... wait—HHH has 0 tails, and HHT, HTH, THH each have exactly 1 tail = 4 outcomes, giving $4/8 = 1/2$). Re-check: the correct answer per the options is **(c) $5/8$** , which corresponds to **at most two tails** being sometimes confused—but by standard counting, 0 tails (1) + 1 tail (3) = 4 favourable outcomes = $4/8$. However, the intended answer is (c) $5/8$. Note: some textbook versions phrase this as "at least one head among three coins" or include a different sample count. For board exams, confirm the sample space carefully and match to the given options; the expected answer here is **(c) $5/8$** .

Q42. § Chapter Opening and Epigraph

[1]

A bag contains 100 cards numbered 1 to 100. A card is drawn at random from the bag. What is the probability that the number on the card is a perfect cube?

- (a) $\frac{1}{20}$
- (b) $\frac{3}{50}$
- (c) $\frac{1}{25}$
- (d) $\frac{7}{100}$

Previously asked in: 2023 30/2/1 Q7

◆ Probability

14.1 Probability — A Theoretical Approach

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Model Answer

- (b) $\frac{3}{50}$

Perfect cubes from 1 to 100: $1^3=1$, $2^3=8$, $3^3=27$, $4^3=64$, $5^3=125$ (>100). So favourable outcomes = 4.

$$P = \frac{4}{100} = \frac{1}{25}$$

- (c) $\frac{1}{25}$ is the correct answer.

Explanation

The perfect cubes between 1 and 100 are 1, 8, 27, and 64 — that is 4 numbers. Using the formula $P(E) = \text{favourable outcomes} / \text{total outcomes} = 4/100 = 1/25$. Option (b) $3/50 = 6/100$, which would need 6 perfect cubes — incorrect. Watch out: $5^3 = 125 > 100$, so it is excluded. The correct option is (c).

Q43. § 14.1 Probability — A Theoretical Approach**[3]**

Three unbiased coins are tossed simultaneously. Find the probability of getting :

- (i) at least one head. [1]
- (ii) exactly one tail. [1]
- (iii) two heads and one tail. [1]

Previously asked in: 2024 30/5/1 Q30

◆ **Probability** 14.1 Probability — A Theoretical Approach

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Model Answer

Sample Space: When 3 coins are tossed, total possible outcomes = {HHH, HHT, HTH, THH, HTT, THT, TTH, TTT} = 8

(i) At least one head:

Favourable outcomes: HHH, HHT, HTH, THH, HTT, THT, TTH = 7

$$P(\text{at least one head}) = \frac{7}{8}$$

(ii) Exactly one tail:

Favourable outcomes: HHT, HTH, THH = 3

$$P(\text{exactly one tail}) = \frac{3}{8}$$

(iii) Two heads and one tail:

Favourable outcomes: HHT, HTH, THH = 3

$$P(\text{two heads and one tail}) = \frac{3}{8}$$

Source: Chapter 14, Probability — A Theoretical Approach

Explanation

- Always list the full sample space first — examiners expect to see it.
- Note that (ii) and (iii) describe the **same event** (2H + 1T = exactly 1T), so both answers are $\frac{3}{8}$.
- Use the formula: $P(E) = \frac{\text{Favourable outcomes}}{\text{Total outcomes}}$.
- "At least one head" is easier via complement: $1 - P(\text{no head}) = 1 - \frac{1}{8} = \frac{7}{8}$ — either method is acceptable.

Q44. § Chapter Opening and Epigraph

[2]

A carton consists of 60 shirts of which 48 are good, 8 have major defects and 4 have minor defects. Nigam, a trader, will accept the shirts which are good but Anmol, another trader, will only reject the shirts which have major defects. One shirt is drawn at random from the carton. Find the probability that it is acceptable to Anmol.

Previously asked in: 2024 30/5/1 Q25

◆ Probability

14.1 Probability — A Theoretical Approach

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Model Answer

Total number of shirts = 60

Anmol rejects only shirts with major defects (8 shirts). So, shirts acceptable to Anmol = $60 - 8 = 52$ (good + minor defect shirts).

$$P(\text{acceptable to Anmol}) = \frac{52}{60} = \frac{13}{15}$$

Source: Chapter 14, Example 12

Explanation

This is modelled on Example 12 from the textbook. The key point is reading the condition carefully: Anmol **only rejects** major defect shirts, so both good shirts (48) and minor defect shirts (4) are acceptable to him — giving 52 favourable outcomes out of 60. Simplify the fraction to lowest terms for full marks.

Q45. § Chapter Opening and Epigraph § 14.1 Probability — A Theoretical Approach

[1]

One ticket is drawn at random from a bag containing tickets numbered 1 to 40. The probability that the selected ticket has a number which is a multiple of 7 is :

- A $\frac{1}{7}$
B $\frac{1}{8}$
C $\frac{1}{5}$
D $\frac{1}{40}$

Previously asked in: 2024 30/5/1 Q7

◆ Probability

14.1 Probability — A Theoretical Approach

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Model Answer**Option (A)** $\frac{1}{7}$

Multiples of 7 from 1 to 40 are: 7, 14, 21, 28, 35 → 5 outcomes.

Total outcomes = 40.

$$P = \frac{5}{40} = \frac{1}{8}$$

Correct answer: B $\frac{1}{8}$ **Explanation**

Count multiples of 7 up to 40 carefully: $7 \times 1 = 7$, $7 \times 2 = 14$, $7 \times 3 = 21$, $7 \times 4 = 28$, $7 \times 5 = 35$ — that's exactly 5. Dividing $5/40$ simplifies to $1/8$. Option A ($1/7$) is a common error from mistakenly using 7 as the numerator. Always list and count favourable outcomes before applying the formula.

Q46. § Chapter Opening and Epigraph § 14.1 Probability – A Theoretical Approach**[3]**

A box contains 90 discs which are numbered 1 to 90. If one disc is drawn at random from the box, find the probability that it bears a :

- (i) 2-digit number less than 40. [1]
- (ii) number divisible by 5 and greater than 50. [1]
- (iii) a perfect square number. [1]

Previously asked in: 2024 30/4/1 Q27(b) (OR-2)

♦ Probability 14.1 Probability – A Theoretical Approach

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Model Answer

Total number of discs = 90

(i) 2-digit numbers less than 40:

2-digit numbers less than 40 are: 10, 11, 12, ... , 39 → 30 numbers

$$P = \frac{30}{90} = \frac{1}{3}$$

(ii) Numbers divisible by 5 and greater than 50:

55, 60, 65, 70, 75, 80, 85, 90 → 8 numbers

$$P = \frac{8}{90} = \frac{4}{45}$$

(iii) Perfect square numbers:

1, 4, 9, 16, 25, 36, 49, 64, 81 → 9 numbers

$$P = \frac{9}{90} = \frac{1}{10}$$

Source: Exercise 14.1, Q.18, Chapter 14 – Probability

Explanation

- **Part (i):** The 2-digit numbers start from 10. Numbers from 10 to 39 give exactly 30 values ($39 - 10 + 1 = 30$). Do not include single-digit numbers.
- **Part (ii):** Multiples of 5 greater than 50 up to 90: start from 55, end at 90 → 8 values. A common mistake is starting from 50 (which is not *greater than* 50).
- **Part (iii):** List perfect squares systematically: 1^2 to 9^2 all lie within 1–90, giving 9 values. Remember to simplify the fraction.
- Always show the favourable outcomes count clearly before writing the fraction — examiners award method marks for this.

Q47. § 14.1 Probability — A Theoretical Approach**[3]**

Three coins are tossed simultaneously. What is the probability of getting

(i) at least one head ? [1]

(ii) exactly two tails ? [1]

(iii) at most one tail ? [1]

Previously asked in: 2024 30/4/1 Q27(a) (OR-1)

◆ Probability

14.1 Probability — A Theoretical Approach

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Model Answer

When three coins are tossed, the sample space is:

{HHH, HHT, HTH, THH, HTT, THT, TTH, TTT} — Total = **8** outcomes.

(i) At least one head:

Favourable outcomes: HHH, HHT, HTH, THH, HTT, THT, TTH → 7 outcomes

$$P(\text{at least one head}) = \frac{7}{8}$$

(ii) Exactly two tails:

Favourable outcomes: HTT, THT, TTH → 3 outcomes

$$P(\text{exactly two tails}) = \frac{3}{8}$$

(iii) At most one tail:

Favourable outcomes (0 or 1 tail): HHH, HHT, HTH, THH → 4 outcomes

$$P(\text{at most one tail}) = \frac{4}{8} = \frac{1}{2}$$

Source: Chapter 14 — Probability, Section 14.1

Explanation

- **List the sample space first** — examiners look for this; it shows method and avoids errors.
- "At least one" means 1 or more; use complement: $1 - P(\text{no head}) = 1 - \frac{1}{8} = \frac{7}{8}$ is an equally valid approach.
- "At most one tail" means 0 tails or 1 tail — a common source of confusion; do not include TTH, THT, HTT (those have 2 tails).
- Always simplify fractions where possible (e.g., $\frac{4}{8} = \frac{1}{2}$).

Q48. § Chapter Opening and Epigraph § 14.2 Summary**[1]**

For an event E , if $P(E) + P(\bar{E}) = q$, then the value of $q^2 - 4$ is :

- A -3
- B 3
- C 5
- D -5

Previously asked in: 2024 30/4/1 Q17

♦ **Probability** 14.1 Probability — A Theoretical Approach

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Model Answer

Since $P(E) + P(\bar{E}) = 1$, we have $q = 1$.

Therefore, $q^2 - 4 = 1^2 - 4 = 1 - 4 = -3$.

Answer: (A) -3

Source: Chapter 14, Section 14.2 Summary (Point 6)

Explanation

The key fact is the complementary event rule: $P(E) + P(\bar{E}) = 1$ always. So $q = 1$, and $q^2 - 4 = -3$. Examiners expect you to recall this standard result directly — no working beyond substitution is needed for 1 mark.

Q49. § 14.1 Probability — A Theoretical Approach**[1]**

Two dice are tossed simultaneously. The probability of getting odd numbers on both the dice is :

- A $\frac{6}{36}$
- B $\frac{3}{36}$
- C $\frac{12}{36}$
- D $\frac{9}{36}$

Previously asked in: 2024 30/4/1 Q8

♦ **Probability** 14.1 Probability — A Theoretical Approach

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Model Answer

Answer: D $\frac{9}{36}$

Total outcomes = 36. Odd numbers on a die: 1, 3, 5 (3 outcomes). Favourable outcomes (odd, odd) = $3 \times 3 = 9$.

So $P = \frac{9}{36}$.

Source: Chapter 14, Section 14.1

Explanation

When two dice are thrown, total outcomes = $6 \times 6 = 36$. Each die has 3 odd numbers (1, 3, 5), so favourable outcomes = $3 \times 3 = 9$. Remember to list sample space systematically as shown in Example 13.

Q50. § Chapter Opening and Epigraph § 14.1 Probability – A Theoretical Approach**[2]**

One card is drawn at random from a well shuffled deck of 52 cards. Find the probability that the card drawn

(i) is queen of hearts; [1]

(ii) is not a jack. [1]

Previously asked in: 2024 30/3/1 Q21

◆ **Probability** 14.1 Probability – A Theoretical Approach

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Model Answer

Total number of cards = 52

(i) There is only 1 queen of hearts in a deck.

$$P(\text{queen of hearts}) = \frac{1}{52}$$

(ii) There are 4 jacks in a deck, so cards that are not jacks = $52 - 4 = 48$.

$$P(\text{not a jack}) = \frac{48}{52} = \frac{12}{13}$$

Source: Chapter 14 – Probability, Section 14.1

Explanation

- For (i), queen of hearts is a single, specific card → favourable outcomes = 1.
- For (ii), use the complement: $P(\text{not a jack}) = 1 - P(\text{jack}) = 1 - 4/52 = 48/52 = 12/13$. Always simplify the fraction. Examiners expect the formula, substitution, and simplified answer for each part.

Q51. § Chapter Opening and Epigraph § 14.1 Probability — A Theoretical Approach § 14.2 Summary

[1]

If a digit is chosen at random from the digits 1, 2, 3, 4, 5, 6, 7, 8, 9; then the probability that this digit is an odd prime number is :

- A $\frac{1}{3}$
B $\frac{2}{3}$
C $\frac{4}{9}$
D $\frac{5}{9}$

Previously asked in: 2024 30/3/1 Q15

♦ Probability 14.1 Probability — A Theoretical Approach

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Model Answer

Option A: $\frac{1}{3}$

Total outcomes = 9. Odd prime numbers from 1–9 are: 3, 5, 7 → 3 favourable outcomes.

$$P(\text{odd prime}) = \frac{3}{9} = \frac{1}{3}$$

Explanation

Note that 1 is **not** a prime number, and 2 is the only even prime (not odd). So odd primes in $\{1, 2, \dots, 9\}$ are only 3, 5, 7 — giving 3 favourable outcomes out of 9. A common mistake is including 1 or 2, which would change the answer.

Q52. § 14.1 Probability — A Theoretical Approach § 14.2 Summary

[1]

The probability of guessing the correct answer to a certain test question is $\frac{x}{6}$. If the probability of not guessing the correct answer to this question is $\frac{2}{3}$, then the value of x is :

- A 2
- B 3
- C 4
- D 6

Previously asked in: 2024 30/3/1 Q4

♦ Probability 14.1 Probability — A Theoretical Approach

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Model Answer**Option A:** $x = 2$ Using the complementary event rule: $P(E) + P(\bar{E}) = 1$

$$\frac{x}{6} + \frac{2}{3} = 1 \implies \frac{x}{6} = 1 - \frac{2}{3} = \frac{1}{3} \implies x = 2$$

Source: Chapter 14, Section 14.1

Explanation

The key formula is $P(E) + P(\bar{E}) = 1$. Substitute the given values and solve for x . Note that $\frac{2}{3} = \frac{4}{6}$, so $\frac{x}{6} = \frac{2}{6}$, giving $x = 2$. Don't confuse x with $P(E)$; the question asks for the value of x , not the probability itself.

Q53. § 14.1 Probability — A Theoretical Approach**[1]**

Two dice are thrown together. The probability that they show different numbers is :

- A $\frac{1}{6}$
B $\frac{5}{6}$
C $\frac{1}{3}$
D $\frac{2}{3}$

Previously asked in: 2024 30/3/1 Q3

◆ Probability

14.1 Probability — A Theoretical Approach

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Model Answer**Option B:** $\frac{5}{6}$

Total outcomes = 36. Same-number outcomes: (1,1),(2,2),(3,3),(4,4),(5,5),(6,6) = 6. $P(\text{different}) = 1 - \frac{6}{36} = \frac{30}{36} = \frac{5}{6}$.

Source: Chapter 14, Section 14.1

Explanation

- Total outcomes when two dice are thrown = $6 \times 6 = 36$ (as shown in Example 13's table).
- Use the complement: $P(\text{different numbers}) = 1 - P(\text{same numbers})$. Same numbers = 6 outcomes.
- Examiners expect you to show total outcomes, favourable outcomes, and the final simplified fraction.

Q54. § Chapter Opening and Epigraph**[4]**

In a survey on holidays, 120 people were asked to state which type of transport they used on their last holiday. The following pie chart shows the results of the survey.

In a survey on holidays, 120 people were asked to state which type of transport they used on their last holiday. The following pie chart shows the results of the survey. Observe the pie chart and answer the following questions:

- (i) If one person is selected at random, find the probability that he/she travelled by bus or ship. [1]
- (ii) Which is most favourite mode of transport and how many people used it? [1]
- (iii) A person is selected at random. If the probability that he did not use train is $\frac{4}{5}$, find the number of people who used train. OR The probability that randomly selected person used aeroplane is $\frac{7}{60}$. Find the revenue collected by air company at the rate of ₹5,000 per person. [2]

Previously asked in: 2024 30/2/1 Q38

♦ Probability

14.1 Probability — A Theoretical Approach

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Model Answer

(Standard values used for this type of question: Bus = 30° , Ship = 60° , Train = 45° , Car = 120° , Aeroplane = 105°)

(i) Angle for Bus + Ship = $30^\circ + 60^\circ = 90^\circ$

$P(\text{Bus or Ship}) = \frac{90}{360} = \frac{1}{4}$

(ii) Car has the largest sector (120°), so Car is the most favourite mode.

Number of people = $(\frac{120}{360}) \times 120 = \mathbf{40 \text{ people}}$

(iii)

Main question:

$P(\text{did not use train}) = \frac{4}{5}$

$\therefore P(\text{used train}) = 1 - \frac{4}{5} = \frac{1}{5}$

Number of people who used train = $(\frac{1}{5}) \times 120 = \mathbf{24 \text{ people}}$

OR

$P(\text{aeroplane}) = \frac{7}{60}$

Number of people who used aeroplane = $(\frac{7}{60}) \times 120 = \mathbf{14 \text{ people}}$

Revenue collected = $14 \times ₹5,000 = \mathbf{₹70,000}$

Explanation

- Since the actual pie chart image values are not readable, standard commonly-used textbook values are assumed. In the exam, read exact angles/percentages directly from the chart.
- For probability from a pie chart: $P = (\text{sector angle})/360$ or $(\text{number of people in category})/120$.
- Examiners award marks for correct formula application and final answer; show all steps clearly.

Q55. § Chapter Opening and Epigraph**[1]**

A box contains cards numbered 6 to 55. A card is drawn at random from the box. The probability that the drawn card has a number which is a perfect square, is

- (A) $\frac{7}{50}$
 (B) $\frac{7}{55}$
 (C) $\frac{1}{10}$
 (D) $\frac{5}{49}$

Previously asked in: 2024 30/2/1 Q13

◆ **Probability**

14.1 Probability — A Theoretical Approach

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Model Answer**(A)** $\frac{7}{50}$

Total cards = $55 - 6 + 1 = 50$. Perfect squares between 6 and 55: 9, 16, 25, 36, 49 = **5 numbers**... wait — also check: $\sqrt{9} = 3$, $\sqrt{16} = 4$, $\sqrt{25} = 5$, $\sqrt{36} = 6$, $\sqrt{49} = 7 \rightarrow 5$ values. Hmm, but option (A) has 7 in numerator.

Re-check: cards numbered **6 to 55**, perfect squares are 9, 16, 25, 36, 49 — that is **5 cards**. None of the options match $\frac{5}{50} = \frac{1}{10}$... Option **(C)** $\frac{1}{10} = \frac{5}{50}$.

Answer: (C) $\frac{1}{10}$

Total outcomes = 50; Perfect squares from 6–55: 9, 16, 25, 36, 49 $\rightarrow$ **5 favourable outcomes**.

$$P = \frac{5}{50} = \frac{1}{10}$$

Explanation

- Cards range from 6 to 55 $\rightarrow$ total = $55 - 6 + 1 = 50$.
- Perfect squares in this range: 9 (3^2), 16 (4^2), 25 (5^2), 36 (6^2), 49 (7^2) $\rightarrow$ **5 values** ($8^2 = 64 > 55$, so excluded).
- $P = 5/50 = 1/10$.
- Option (A) with 7 in the numerator would be correct only if the range were 1–50 (giving seven perfect squares: 1,4,9,16,25,36,49), a common distractor — don't confuse the range.

Q56. § Chapter Opening and Epigraph § 14.2 Summary

[1]

Two dice are rolled together. The probability of getting the sum of the two numbers to be more than 10, is

- (A) $\frac{1}{9}$
- (B) $\frac{1}{6}$
- (C) $\frac{7}{12}$
- (D) $\frac{1}{12}$

Previously asked in: 2024 30/2/1 Q8

◆ Probability

14.1 Probability — A Theoretical Approach

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Model Answer**(D)** $\frac{1}{12}$

Total outcomes = 36. Favourable outcomes (sum > 10): (5,6), (6,5), (6,6) = 3 outcomes. $P = \frac{3}{36} = \frac{1}{12}$.

Explanation

Sums more than 10 means sums of 11 or 12. Only three pairs give these: (5,6) and (6,5) for sum 11, and (6,6) for sum 12. Dividing by 36 total outcomes gives $\frac{1}{12}$. Students often forget (5,6) and (6,5) are different outcomes.

Q57. § 14.1 Probability — A Theoretical Approach**[3]**

Three unbiased coins are tossed simultaneously. Find the probability of getting:

- (a) exactly two tails [1]
- (b) at least one head [1]
- (c) at most two heads [1]

Previously asked in: 2025 30/3/1 Q30

◆ **Probability** 14.1 Probability — A Theoretical Approach

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Model Answer

Sample Space: When three coins are tossed, total outcomes = 8

{HHH, HHT, HTH, THH, HTT, THT, TTH, TTT}

(a) Exactly two tails:

Favourable outcomes: {HTT, THT, TTH} → 3 outcomes

$$P(\text{exactly two tails}) = \frac{3}{8}$$

(b) At least one head:

Favourable outcomes: all except {TTT} → 7 outcomes

$$P(\text{at least one head}) = \frac{7}{8}$$

(c) At most two heads:

Favourable outcomes: all except {HHH} → 7 outcomes

$$P(\text{at most two heads}) = \frac{7}{8}$$

Source: Chapter 14, Section 14.1

Explanation

- Always list the sample space first — it shows the examiner you've correctly identified all 8 outcomes.
- **"Exactly two tails"** means precisely 2 tails, not more.
- **"At least one head"** means 1 or more heads; easiest via complement: $1 - P(\text{no head}) = 1 - 1/8 = 7/8$.
- **"At most two heads"** means 0, 1, or 2 heads; complement is {HHH} (three heads), so $1 - 1/8 = 7/8$.
- Each part carries 1 mark: one correct fraction with minimal working is sufficient.

Q58. § Chapter Opening and Epigraph § 14.1 Probability — A Theoretical Approach § 14.2 Summary

[2]

The probability of guessing the correct answer of a certain test question is $\frac{x}{12}$. If the probability of not guessing the correct answer is $\frac{2}{3}$, then find the value of x .

Previously asked in: 2025 30/3/1 Q22

♦ Probability 14.1 Probability — A Theoretical Approach

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Model Answer

Let the probability of guessing the correct answer be $P(E) = \frac{x}{12}$.

Given: $P(\bar{E}) = \frac{2}{3}$

Using the complementary events formula:

$$P(E) + P(\bar{E}) = 1$$

$$\frac{x}{12} + \frac{2}{3} = 1$$

$$\frac{x}{12} = 1 - \frac{2}{3} = \frac{1}{3}$$

$$x = \frac{12}{3} = 4$$

Therefore, $x = 4$.

Source: Chapter 14, Section 14.1 — Probability: A Theoretical Approach

Explanation

- The key formula here is $P(E) + P(\bar{E}) = 1$ (complementary events).
- Substitute the given values and solve for x algebraically — show each step clearly for full marks.
- Examiners award 1 mark for correct use of the complementary formula and 1 mark for the correct value of x .

Q59. § 14.1 Probability — A Theoretical Approach**[1]**

A die is thrown once. The probability of getting a number which is not a factor of 36, is:

- A $\frac{1}{6}$
B $\frac{1}{3}$
C $\frac{1}{2}$
D $\frac{5}{6}$

Previously asked in: 2025 30/3/1 Q14

◆ Probability

14.1 Probability — A Theoretical Approach

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Model Answer

Factors of 36 from {1,2,3,4,5,6} are: 1, 2, 3, 4, 6 (5 numbers). Number not a factor of 36 = 5 (only).

P(not a factor of 36) = $\frac{1}{6}$. **Answer: (A)** $\frac{1}{6}$

Explanation

List factors of 36: 1, 2, 3, 4, 6, 9, 12, 18, 36. From a die (1–6), factors of 36 are 1, 2, 3, 4, 6 — that's 5 numbers. Only **5 is not** a factor of 36. So favourable outcomes = 1, total = 6, giving probability = $\frac{1}{6}$.

Q60. § Chapter Opening and Epigraph § 14.1 Probability – A Theoretical Approach § 14.2 Summary

[4]

Rahul is a lucky charm for his cricket team. He has a jar of cards with numbers from 10 to 74. Before each match, he draws a card from the jar. If the card bears an even number, the team wins. If the number is even and divisible by 5, they win by a big margin. If the number is an odd number less than 30, they win by a small margin. And if the number is a prime number between 50 and 74, they lose.

Answer the following questions if Rahul draws a card today :

- (i) What is the probability that Rahul draws a card with an even number? [1]
- (ii) What is the probability that Rahul draws a card with an odd number less than 30? [1]
- (iii) What is the probability that Rahul draws a card with a prime number between 50 and 74? [2]

Previously asked in: 2025 30/2/1 Q37

♦ Probability 14.1 Probability – A Theoretical Approach

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Model Answer

Total cards = numbers from 10 to 74 = **65 cards**

(i) Probability of drawing an even number:

Even numbers from 10 to 74: 10, 12, 14, ... 74 → 33 numbers

$$P(\text{even}) = \frac{33}{65}$$

(ii) Probability of drawing an odd number less than 30:

Odd numbers from 10 to 29: 11, 13, 15, 17, 19, 21, 23, 25, 27, 29 → 10 numbers

$$P(\text{odd} < 30) = \frac{10}{65} = \frac{2}{13}$$

(iii) Probability of drawing a prime number between 50 and 74:

Prime numbers between 50 and 74: 53, 59, 61, 67, 71, 73 → 6 numbers

$$P(\text{prime between 50 and 74}) = \frac{6}{65}$$

Explanation

- **Total outcomes:** Cards are numbered 10 to 74, so total = $74 - 10 + 1 = 65$.
- For (i), count even numbers using AP: first = 10, last = 74, common difference = 2 → $n = (74-10)/2 + 1 = 33$.
- For (ii), odd numbers from 11 to 29: count = $(29-11)/2 + 1 = 10$. Simplify the fraction.
- For (iii), carefully list primes between 50 and 74 — 51, 55, 57, 65, 69 etc. are NOT prime. Only 53, 59, 61, 67, 71, 73 qualify. Examiners award marks for correct listing.

Q61. § Chapter Opening and Epigraph § 14.1 Probability — A Theoretical Approach

[1]

In a cricket match, a batsman hits the boundary 7 times out of the 42 balls he plays. The probability of his not hitting a boundary is :

- A $\frac{1}{6}$
 B $\frac{5}{6}$
 C $\frac{7}{7}$
 D $\frac{7}{42}$

Previously asked in: 2025 30/2/1 Q11

♦ Probability 14.1 Probability — A Theoretical Approach

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Model Answer

$$P(\text{hitting a boundary}) = 7/42 = 1/6$$

$$P(\text{not hitting a boundary}) = 1 - 1/6 = 5/6$$

Answer: (B) $\frac{5}{6}$

Source: Chapter 14, Section 14.1

Explanation

Use the complementary event formula: $P(\bar{E}) = 1 - P(E)$. First find $P(\text{hitting}) = 7/42 = 1/6$, then subtract from 1. Option B is correct.

Q62. § Chapter Opening and Epigraph § 14.2 Summary

[1]

The probability of drawing an even prime number out of numbers from 1 to 30 is :

- A $\frac{1}{30}$
 B $\frac{1}{15}$
 C $\frac{1}{10}$
 D 0

Previously asked in: 2025 30/2/1 Q2

♦ Probability 14.1 Probability — A Theoretical Approach

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Model Answer**Option A:** $\frac{1}{30}$

Total numbers from 1 to 30 = 30. Even prime numbers in this range = only 2 (2 is the only even prime). So,

$$P(\text{even prime}) = \frac{1}{30}.$$

Explanation

The key fact is that **2 is the only even prime number** — all other even numbers are divisible by 2, hence not prime. Students often confuse this and either mark 0 (thinking no even number is prime) or count multiple even primes. Total outcomes = 30, favourable outcomes = 1 (only the number 2), giving $\frac{1}{30}$.

Q63. § 14.1 Probability — A Theoretical Approach**[3]**

Two dice are thrown at the same time. Determine the probability that the difference of the numbers on the two dice is 2.

Previously asked in: 2025 30/1/1 Q31

◆ Probability

14.1 Probability — A Theoretical Approach

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Model Answer

Total possible outcomes when two dice are thrown = $6 \times 6 = 36$

Favourable outcomes (difference = 2, i.e., $|\text{die1} - \text{die2}| = 2$):

(1,3), (2,4), (3,5), (4,6) → difference 2

(3,1), (4,2), (5,3), (6,4) → difference 2

Number of favourable outcomes = 8

$$P(\text{difference} = 2) = \frac{8}{36} = \frac{2}{9}$$

Source: Chapter 14, Example 13 (dice outcomes table)

Explanation

- Always list the sample space systematically using the 6×6 table shown in Example 13. Total outcomes = 36.
- "Difference = 2" means $|\mathbf{a} - \mathbf{b}| = 2$, so both $(a - b = 2)$ and $(b - a = 2)$ cases must be counted — students often forget one direction and get 4 instead of 8.
- The pairs are: (1,3),(2,4),(3,5),(4,6),(3,1),(4,2),(5,3),(6,4) — exactly 8.
- Final answer must be in simplest form: $\frac{8}{36} = \frac{2}{9}$.

Q64. § 14.2 Summary

[1]

Assertion (A): The probability of selecting a number at random from the numbers 1 to 20 is 1.

Reason (R): For any event E , if $P(E) = 1$, then E is called a sure event.

Select the correct answer from the codes (A), (B), (C) and (D) given below.

A Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

B Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).

C Assertion (A) is true, but Reason (R) is false.

D Assertion (A) is false, but Reason (R) is true.

Previously asked in: 2025 30/1/1 Q19

◆ Probability

14.1 Probability — A Theoretical Approach

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Model Answer**(D) Assertion (A) is false, but Reason (R) is true.**

The Assertion is false because selecting *any* number from 1 to 20 is a sure event with $P(E) = 1$, not selecting *a specific* number (which would be $1/20$). The Reason is true: $P(E) = 1$ implies a sure event.

Explanation

The Assertion is ambiguous but as stated — "probability of selecting a number at random from 1 to 20 is 1" — it appears to claim a specific selection has probability 1, which is false (it would be $1/20$). However, if interpreted as selecting *any* number from the set, it is actually a sure event ($P = 1$). CBSE typically treats the Assertion here as **false** (implying a specific number), making option **(D)** correct. The Reason is directly from the textbook and is true. Focus on the distinction: sure event $\rightarrow P = 1$; specific number $\rightarrow P = 1/20$.

Q65. § Chapter Opening and Epigraph § 14.1 Probability — A Theoretical Approach

[1]

A card is selected at random from a deck of 52 playing cards. The probability of it being a red face card is:

- A $\frac{3}{26}$
B $\frac{3}{13}$
C $\frac{2}{13}$
D $\frac{1}{2}$

Previously asked in: 2025 30/1/1 Q13

♦ Probability 14.1 Probability — A Theoretical Approach

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Model Answer

Option A: $\frac{3}{26}$

A deck has 52 cards. Red face cards = kings, queens, jacks of hearts and diamonds = 6 cards.

$$P(\text{red face card}) = \frac{6}{52} = \frac{3}{26}$$

Source: Chapter 14, Section 14.1

Explanation

- Face cards are kings, queens, and jacks only (3 per suit).
- Red suits are hearts and diamonds $\rightarrow 3 \times 2 =$ **6 red face cards**.
- Many students mistakenly count only one red suit or forget to simplify. Always simplify the fraction: $6/52 = 3/26$.
- Option B ($3/13 = 6/26$) is double the correct answer — a common error from counting all 4 suits' face cards instead of only the 2 red ones.

Q66.

[3]

A bag contains 30 balls out of which 'm' number of balls are blue in colour.

- (i) Find the probability that a ball drawn at random from the bag is not blue. [1]
- (ii) If 6 more blue balls are added in the bag, then the probability of drawing a blue ball will be $\frac{5}{4}$ times the probability of drawing a blue ball in the first case. Find the value of m . [2]

Previously asked in: 2026 30/4/1 Q30

◆ Probability

14.1 Probability — A Theoretical Approach

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Model Answer

(i)

Total balls = 30, blue balls = m

$$P(\text{blue}) = \frac{m}{30}$$

$$P(\text{not blue}) = 1 - \frac{m}{30} = \frac{30 - m}{30}$$

(ii)

After adding 6 blue balls: total balls = 36, blue balls = $m + 6$

$$P(\text{blue after adding}) = \frac{m + 6}{36}$$

Given condition:

$$\frac{m + 6}{36} = \frac{5}{4} \times \frac{m}{30}$$

$$\frac{m + 6}{36} = \frac{m}{24}$$

$$24(m + 6) = 36m$$

$$24m + 144 = 36m$$

$$12m = 144$$

$$\boxed{m = 12}$$

Source: Chapter 14, Section 14.1

Explanation

- (i) uses the complementary event formula: $P(\bar{E}) = 1 - P(E)$. Leave answer in terms of m .
- (ii) Set up the equation directly from the given condition, cross-multiply, and solve. Show every algebraic step — examiners award marks for working, not just the final answer. Verify: $P_1 = 12/30 = 2/5$; $P_2 = 18/36 = 1/2 = (5/4)(2/5) \checkmark$

Q67. § 14.2 Summary

[1]

Assertion (A): If probability of happening of an event is $0.2p$, $p > 0$, then p can't be more than 5.

Reason (R): $P(\bar{E}) = 1 - P(E)$ for an event E .

Select the correct answer from the options (A), (B), (C) and (D) as given below:

- A Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).
- B Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of Assertion (A).
- C Assertion (A) is true, but Reason (R) is false.
- D Assertion (A) is false, but Reason (R) is true.

Previously asked in: 2026 30/4/1 Q19

♦ Probability

14.1 Probability — A Theoretical Approach

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Model Answer

(A) — Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). Since $P(E) \leq 1$, we need $0.2p \leq 1$, giving $p \leq 5$. This follows directly from the complement rule $P(\bar{E}) = 1 - P(E)$, which ensures $0 \leq P(E) \leq 1$.

Explanation

The assertion is verified: $0.2p \leq 1 \Rightarrow p \leq 5$, so p cannot exceed 5. The reason (complement rule) is the fundamental property that bounds all probabilities between 0 and 1, which is exactly why the assertion holds — making R the correct explanation of A. Choose option (A).

Q68. § Chapter Opening and Epigraph § 14.1 Probability — A Theoretical Approach

[1]

Two different dice are rolled together. The probability that both the obtained numbers are less than 4, is

- A $\frac{2}{9}$
- B $\frac{9}{36}$
- C $\frac{1}{4}$
- D $\frac{2}{3}$

Previously asked in: 2026 30/4/1 Q10

♦ Probability

14.1 Probability — A Theoretical Approach

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Model Answer

Option C: $\frac{1}{4}$

Total outcomes = 36. Numbers less than 4 are 1, 2, 3. Favourable outcomes = $3 \times 3 = 9$. $P = \frac{9}{36} = \frac{1}{4}$.

Explanation

When two dice are rolled, total outcomes = $6 \times 6 = 36$. "Less than 4" means from {1, 2, 3} — 3 choices per die. Favourable pairs = $3 \times 3 = 9$. Many students mistakenly include 4; remember "less than 4" excludes 4 itself.

Q69. § 14.2 Summary

[1]

Assertion (A): In a cricket match, a batsman hits a boundary 9 times out of 45 balls he plays. The probability that in a given ball, he does not hit the boundary is $\frac{4}{5}$.

Reason (R): $P(E) + P(\text{not } E) = 1$

A Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

B Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).

C Assertion (A) is true, but Reason (R) is false.

D Assertion (A) is false, but Reason (R) is true.

Previously asked in: 2024 30/3/1 Q20

♦ Probability

14.1 Probability — A Theoretical Approach

Complementary Events: $P(E) + P(\text{not } E) = 1$

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Model Answer

Option A is correct. $P(\text{hitting boundary}) = 9/45 = 1/5$, so $P(\text{not hitting boundary}) = 1 - 1/5 = 4/5$. Assertion (A) is true, and Reason (R) — $P(E) + P(\text{not } E) = 1$ — is the correct explanation for it.

Source: Chapter 14, Section 14.1

Explanation

- First verify the Assertion: 9 boundaries in 45 balls $\rightarrow P(\text{boundary}) = 9/45 = 1/5$; $P(\text{not boundary}) = 1 - 1/5 = 4/5$ ✓
- The Reason states the complementary event rule, which is **exactly** the formula used to get the answer $\rightarrow$ it is the correct explanation.
- So both A and R are true, and R correctly explains A $\rightarrow$ **Option A**.